

Transmission lines - I

1.1 Introduction:

The electrical lines which are used to transmit the electrical energy along them from one point to another are called Transmission line.

Generally, a transmission line is a distributed parameter network, where the length of the line is much longer than the transmitted signal wavelength.

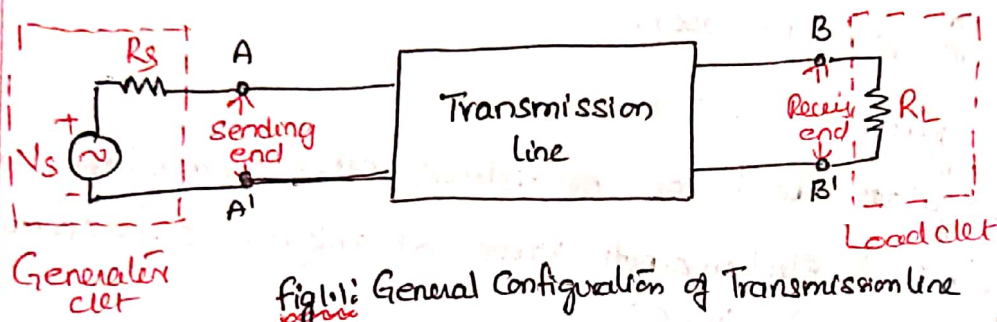


Fig. 1.1: General Configuration of Transmission line

1.2 Types of Transmission line: Depending on the structure and nature, there are various types of Transmission lines.

1. Parallel-wire or Open wire line:

- * These lines are the parallel conductors open to air, and the conductors are separated by air as dielectric and mounted on the posts and towers as shown in fig 1.2(a)
- * Telephone ~~wires~~ lines, telegraphy lines, and power lines are the examples of the open wire lines.
- * These lines are simple, low cost and easy to maintain for short distances.
- * Best suited for low frequency short and medium distance transmission.
- * The use of open wire lines became undesirable above the frequencies of order of 100 MHz, because of radiation losses which increase as frequency increases.

2. Coaxial cables: (Fig 1.2b)

- * The coaxial cable consists of two ~~Conductors~~ ^{Concentric} conductors, separated by a dielectric material, may air, gas or solid.
- * They are used as TV cables and telephone cables.
- * They have less attenuation, negligible radiation compared to open wire lines.
- * These cables are used up to a frequency of 1 GHz, above this frequency, losses in the dielectric increases as frequency increases.

3. Wave guides:

- * Waveguides are hollow or dielectric filled conductors used to transmit electromagnetic waves at microwave frequencies as shown in Fig (1.2c).
- * They are single conductor transmission lines, either rectangular or circular cross section.
- * They have higher power handling capacity than coaxial lines.
- * Power loss is negligible due to perfectly conducting surfaces.
- * They are commonly used as communication networks, circuit components and matching devices at RF frequencies.

4. Optical fibres:

- * Optical fibres are non-conducting circular waveguides made up of silicon oxide or silica as shown in Fig 1.2d.
- * They are long thin wires with a core dielectric (R_1, ϵ_1, n_1) and coated with cladding dielectric of R_2, ϵ_2, n_2 ($n_2 < n_1$).
- * Light waves are transmitted by at 10^{15} GHz frequencies.
- * They are small size, light weight, low losses and very high information carrying capacity when compared to Co-axial cables. Mostly used as communication networks.

5. Microstrip lines

- * A parallel plate transmission line consisting of a conducting strip and a ground plane separated by a dielectric substrate.
- * It is fabricated using PCB technology as shown in fig 1.2 e)
- * The fields are confined to in the dielectric substrate between the strip and the ground plane.
- * These are lighter, more compact and flexible than other type of transmission lines and have low power handling capacity.
- * Used to transmit microwave signals and can be used as microwave components such as antennas, directional couplers, filters etc.

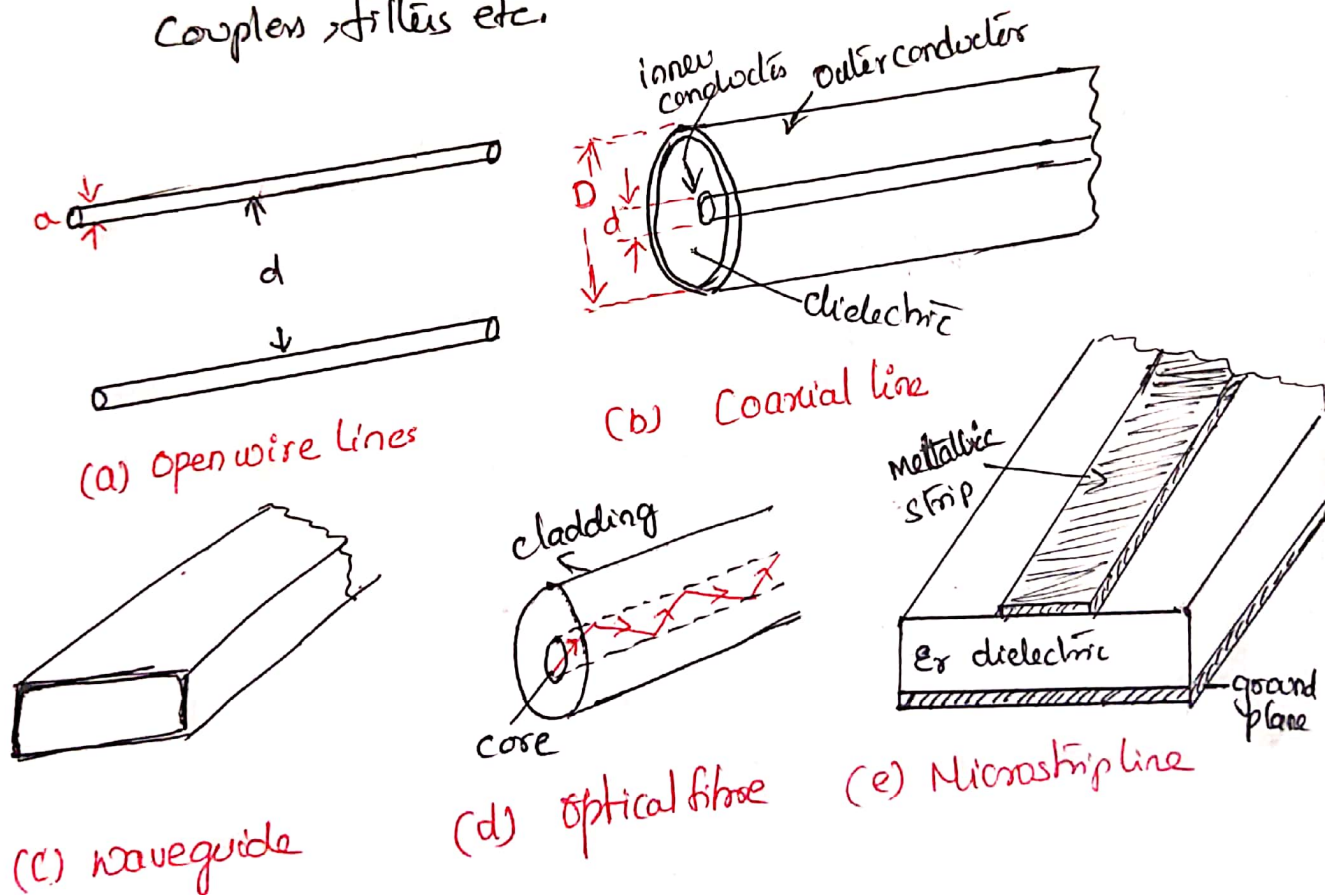


Fig 1.2 Types of Transmission Lines

1.3 Applications of Transmission Lines

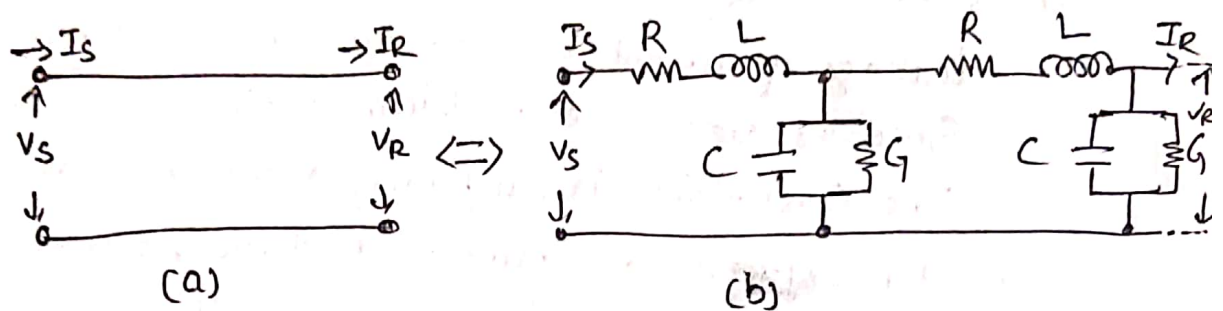
(u)

1. They are used to transfer energy from one circuit to another
2. They can be used as circuit elements like inductors, capacitors and so on.
3. They can be used as stubs, impedance matching devices, measuring devices, etc.
4. They can be used to connect between Televisions, and TV antennas, computer networks, telephony, and power transmission.
5. Coaxial cables are frequently used in Laboratories, a

1.4 Transmission line parameters

(5)

The Equivalent circuit of a two-wire T/L is shown in fig 1.5 illustrating its distributed parameters. It consists of cascaded sections of series resistance R , series Inductance L , shunt capacitance C and shunt conductance G along the length. These parameters are calculated per unit length of the T/L. The performance of transmission line depends on R, L, C & G parameters, called line parameters.



Fig(1.5) : (a) 2-wire T/L (b) Equivalent ckt. of T/L

1. Resistance (R): The series resistance of a T/L is uniformly distributed throughout the length of the line. Each conductor has internal resistance, depends on conductivity (σ) and cross sectional area (a) of the conductors. ($R = \rho / a$).

It is measured as loop resistance per unit length of the line. Its units are Ω/m or Ω/km .

2. Inductance (L): When the conductors of T/L carry some currents, the magnetic flux linkages per unit current give the inductance of a line. It is uniformly distributed along the line. $[L = \frac{N\Phi}{I} = \Phi/I]$

It is measured as loop inductance per unit length. Its units are Henry/m or H/m

3. Capacitance (C) :- The Shunt Capacitance is formed due to electric field between the parallel conductors of a T/L, separated by an insulating medium. The capacitance is uniformly distributed along the length of the line. (6)

$$C = \frac{\epsilon a}{d} = \frac{q}{V}$$

- * It is expressed as shunt capacitance per unit length.
- * Its units are faraads/m or F/m or F/km

4. Conductance (G) :- A Shunt Conductance is due to leakage of current between the conductors of a T/L, since the dielectric or insulation medium is not perfect. The conductance is uniformly distributed along the line.

- * It is expressed as shunt conductance per unit length.
- * Its units are mhos/m or S/m or S/km

* 1.5 Primary constants of Transmission lines

The parameters R, L, C and G are called primary constants of the transmission line. They are assumed to be ~~independent~~ independent of frequency.

$R \rightarrow$ series resistance per unit length of a line (Ω/km)

$L \rightarrow$ series inductance per unit length of a line (H/km)

$C \rightarrow$ shunt capacitance per unit length of a line (F/km)

$G \rightarrow$ shunt conductance per unit length of a line (S/km)

The Series Impedance ' Z ' and Shunt admittance ' Y ' of a line per unit length can be expressed as

$$Z = R + j\omega L$$

$$Y = G + j\omega C$$

The primary constants of different types of T/L are discussed. (7)

i) For Parallel wires:-

$$R_{dc} = \frac{2}{\sigma_c \pi a^2}, \Omega/m \quad (f < 10 \text{ kHz})$$

$$R_{ac} = \sqrt{\frac{f \mu_c}{\pi \sigma_c a^2}} = \sqrt{\frac{R_{dc} f \mu_c}{2}}, \Omega/m \quad (f > 10 \text{ kHz})$$

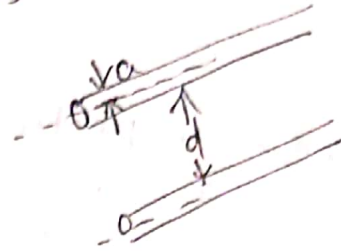
$$L = [4\pi + 9.21 \log_{10} \frac{d}{a}] \times 10^{-7}, \text{ H/m}$$

At high frequencies

$$L = \frac{40}{\pi} \ln\left(\frac{d}{a}\right), \text{ H/m}$$

$$C = \frac{12.07 \epsilon_r}{\ln(d/a)}, \text{ F/m}$$

$$G = \frac{C \sigma}{\epsilon} = \frac{\pi \sigma}{\ln(d/a)}, \text{ S/m}$$



Where σ_c = Conductivity of a conductor

a = radius of wire

d = spacing between the wires ($d > 2a$)

μ_c = permeability of the conductor

ϵ_r = ~~dielectric~~ dielectric constant of the medium

ϵ_r = dielectric constant of the medium

μ_r = relative permeability of the medium ($\mu_r = 1$ for air)

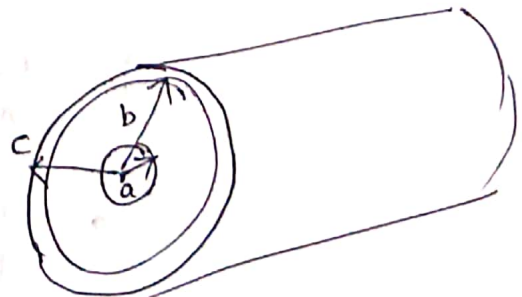
ii) For coaxial cables

$$R_{dc} = \frac{1}{\pi \sigma_c} \left\{ \frac{1}{a^2} + \frac{1}{c^2 - b^2} \right\} \Omega/m$$

$$R_{ac} = \sqrt{\frac{f \mu_c}{4 \pi \sigma_c}} \left(\frac{1}{a} + \frac{1}{b} \right) \Omega/m$$

$$L = \frac{\mu}{2\pi} \ln(b/a); \text{ H/m}$$

$$C = \frac{2\pi \epsilon}{\ln(b/a)}, \text{ F/m}$$



$$G = \frac{C \sigma}{\epsilon} = \frac{2\pi \sigma}{\ln(b/a)}, \text{ S/m}$$

where

(8)

a = radius of the inner conductor

b = inner radius of outer conductor

c = outer radius of outer conductor

σ_c = conductivity of the conductor

f = frequency

μ_c = permeability of the conductor

Example 1: A parallel open line made of aluminium has conductivity $3.6 \times 10^7 \text{ S/m}$ and is kept in air. The conductor radius is 1 mm , separation between the conductors is 0.15 m , operating freq. is 1 kHz . Find the line parameters R, L, C, G .

Solution: Given $\sigma_c = 3.6 \times 10^7 \text{ S/m}$,
 $\mu_c = \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$
 $f = 1 \text{ kHz}$
 $a = 1 \text{ mm} = 0.001 \text{ m}$
 $d = 0.15 \text{ m}$

for air, $\sigma = 0$, $\epsilon = \epsilon_0$, $\mu = \mu_0$

$\therefore$ for open wire,

$$R = \sqrt{\frac{f \mu_c}{\pi \sigma_c a^2}} \Omega/\text{m} = \sqrt{\frac{1 \times 10^3 \times 4\pi \times 10^{-7}}{\pi \times 3.6 \times 10^7 \times (0.001)^2}} = 3.33 \text{ m}\Omega/\text{m}$$

$$L = \frac{\mu_0}{\pi} \ln(d/a) = \frac{4\pi \times 10^{-7}}{\pi} \ln\left(\frac{0.15}{0.001}\right) = 24 \text{ H/m}$$

$$C = \frac{\pi \epsilon}{\ln(d/a)} = \frac{\pi \epsilon_0}{\ln\left(\frac{0.15}{0.001}\right)} = 5.55 \text{ pF/m}$$

$$G = \frac{C \sigma}{\epsilon} = \frac{\pi \sigma}{\ln(d/a)} = 0 \text{ S/m}$$

1.6 Transmission Line equations

(9)

Consider a section of transmission line of length dx . Let R, L, C and G be the primary constants assume these values do not vary with frequency. Consider a point P on the line at a distance x from the source as shown in fig 1.6. Let another point Q be at small distance dx from point P .

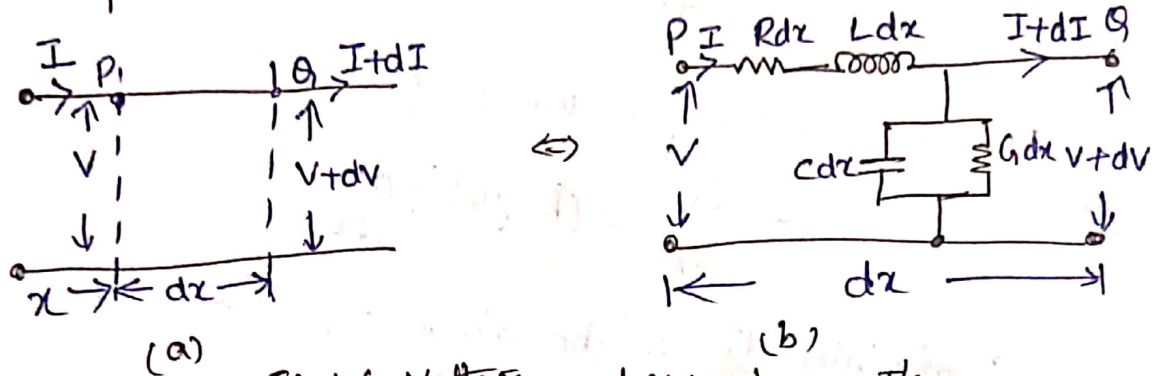


Fig 1.6 Voltages and currents on a T/L

- * Let V and I be the voltage and current resp. at point P , and $V+dv$ and $I+dI$ are voltage and current at point Q .
- * For a small length dx of the line, the series impedance is $(R+j\omega L)dx$ and shunt admittance is $(G+j\omega C)dx$.
- * The voltage drop across series impedance is the Potential difference between P & Q is

$$V - (V+dv) = I(R+j\omega L)dx$$

$$-dv = I(R+j\omega L)dx$$

$\Rightarrow$

$$-\frac{dv}{dx} = (R+j\omega L)I \quad \text{--- (1)}$$

$\Rightarrow$

and

10

The current difference b/w P & Q is the voltage drop across the shunt admittance is

$$\begin{aligned} I - (I + dI) &= V(G + j\omega C) dx \\ \Rightarrow -dI &= V(G + j\omega C) dx \\ \Rightarrow -\frac{dI}{dx} &= (G + j\omega C)V \quad \text{--- (2)} \end{aligned}$$

Differentiating eq (1) w.r.t 'x'

$$-\frac{d^2V}{dx^2} = (R + j\omega L) \frac{dI}{dx}$$

Substituting $\frac{dI}{dx}$ in the above equation

$$\frac{d^2V}{dx^2} = (R + j\omega L)(G + j\omega C)V \quad \text{--- (3)}$$

Differentiating eq (2) w.r.t 'x'

$$-\frac{d^2I}{dx^2} = (G + j\omega C) \frac{dV}{dx}$$

Substituting $\frac{dV}{dx}$ in the above equation

$$\frac{d^2I}{dx^2} = (R + j\omega L)(G + j\omega C)I \quad \text{--- (4)}$$

Let the constant term at given frequency be

$$(R + j\omega L)(G + j\omega C) = \gamma^2$$

$$\text{or } \gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \alpha + j\beta = \sqrt{ZY} \quad \text{--- (5)}$$

where γ = propagation constant of the T/L at given freq.
 α = attenuation constant
 β = phase shift constant

$$\therefore \frac{d^2 V}{dx^2} = \gamma^2 V \quad \text{--- (6)}$$

(11)

$$\text{and } \frac{d^2 I}{dx^2} = \gamma^2 I \quad \text{--- (7)}$$

Eq (6) and (7) are called linear differential equations of the T/L. The solutions of the above equations are

$$V = ae^{\gamma x} + b e^{-\gamma x} \quad \text{--- (8)}$$

$$I = ce^{\gamma x} + d e^{-\gamma x} \quad \text{--- (9)}$$

where a, b, c and d are the arbitrary constants

Eq (8) & (9) represents a standing wave pattern (i.e. sum of forward wave and backward waves).

* The solutions of eq (6) & (7) in terms of hyperbolic functions are given by

* To find the relationship b/w V and I , substitute eq (8) into eq (1)

$$-\frac{d}{dx} (ae^{\gamma x} + b e^{-\gamma x}) = (R + j\omega L) I$$

$$\Rightarrow - [\gamma a e^{\gamma x} - \gamma b e^{-\gamma x}] = (R + j\omega L) I$$

$$\Rightarrow [\gamma b e^{-\gamma x} - \gamma a e^{\gamma x}] = (R + j\omega L) I$$

$$\Rightarrow \gamma [b e^{-\gamma x} - a e^{\gamma x}] = (R + j\omega L) I$$

$$\Rightarrow I = \frac{\gamma [b e^{-\gamma x} - a e^{\gamma x}]}{R + j\omega L} \quad \text{from eq (5)}$$

$$\Rightarrow I = \frac{\sqrt{(R + j\omega L)(G + j\omega C)} [b e^{-\gamma x} - a e^{\gamma x}]}{(R + j\omega L)}$$

$$\Rightarrow I = \frac{\sqrt{\frac{G+j\omega C}{R+j\omega L}} (be^{-\gamma x} - ae^{\gamma x})}{\quad} \quad (12)$$

$$\therefore \Rightarrow I = \frac{1}{Z_0} (be^{-\gamma x} - ae^{\gamma x}) \quad \text{--- (10)}$$

where $Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}} = \sqrt{\frac{Z}{Y}}$ is a constant at a given frequency, called characteristic impedance.

$\therefore$ eq (8) and (9) can be written as

$$\boxed{\begin{aligned} V &= ae^{\gamma x} + be^{-\gamma x} \quad \text{--- (11)} \\ I &= \frac{1}{Z_0} (be^{-\gamma x} - ae^{\gamma x}) \quad \text{--- (12)} \end{aligned}} \quad \text{T/L Equations.}$$

* The solutions of eq (6) & (7) in hyperbolic functions form is given by substituting $e^{\gamma x} = \cosh \gamma x + \sinh \gamma x$, $e^{-\gamma x} = \cosh \gamma x - \sinh \gamma x$

$$V = A \cosh \gamma x + B \sinh \gamma x \quad \text{--- (13)}$$

$$I = -\frac{1}{Z_0} (A \sinh \gamma x + B \cosh \gamma x) \quad \text{--- (14)}$$

A & B are obtained by applying initial conditions.

Let V_s & I_s be the source voltage and source current resp. At the sending end,
 $x=0$, $V=V_s$, $I=I_s$.

$$\therefore V_s = A \cosh \gamma(0) + B \sinh \gamma(0) = A + 0 = A$$

$$\therefore \boxed{V_s = A}$$

$$I_s = -\frac{1}{Z_0} (A \sinh \gamma(0) + B \cosh \gamma(0)) = -\frac{1}{Z_0} (B)$$

$$\Rightarrow \boxed{B = -I_s Z_0}$$

Substituting the constants A and B in eq (13) & (14) (13)

$$\text{and } \boxed{\begin{aligned} V &= V_s \cosh \gamma x - I_s Z_0 \sinh \gamma x & \text{--- (15)} \\ I &= I_s \cosh \gamma x - \frac{V_s}{Z_0} \sinh \gamma x & \text{--- (16)} \end{aligned}}$$

Eq's (15) & (16) are called Transmission line equations, and give Voltage and Current at a point of distance x from the sending end in terms of source voltage and current.

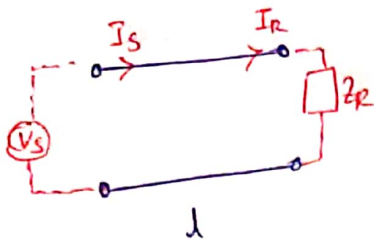
NOTE : γ and Z_0 are called Secondary constants of a Transmission line.

— x —

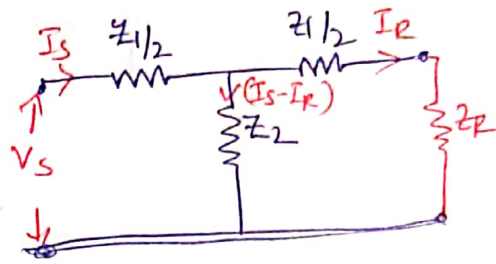
1.7 T & π Equivalent circuit of a T/L

(14)

Consider a short section of transmission, the equivalent T-network is shown in fig(a) & (b)



(a) short section of T/L



(b) Equivalent of a T/L

From transmission line equation

$$I = I_s \cosh \gamma x - \frac{V_s}{Z_0} \sinh \gamma x \quad \text{--- (1)}$$

If Z_0 is the characteristic impedance of a T/L of length l , and γ is the propagation constant, then eq (1) at $x=l$, $I = I_R$

$$I_R = I_s \cosh \gamma l - \frac{V_s}{Z_0} \sinh \gamma l \quad \text{--- (2)}$$

from symmetrical T-Network as shown in fig(b), we see that,

$$V_s = I_s \left(\frac{Z_1}{2} \right) + (I_s - I_R) Z_2$$

$$\text{so that } V_s = I_s \left(\frac{Z_1}{2} + Z_2 \right) - I_R Z_2$$

$$\Rightarrow I_R Z_2 = I_s \left(\frac{Z_1}{2} + Z_2 \right) - V_s$$

$$\Rightarrow I_R = I_s \left(\frac{Z_1}{2Z_2} + 1 \right) - \frac{V_s}{Z_2} \quad \text{--- (3)}$$

For equivalence between T/L and T-network, equating co-efficients of V_s and I_s [from eq (2) & (3)], we have

$$\frac{Z_1}{2Z_2} + 1 = \cosh \gamma l \quad \text{--- (4)}$$

$$\text{and } \frac{1}{Z_2} = \frac{\sinh \gamma l}{Z_0} \Rightarrow \boxed{Z_2 = \frac{Z_0}{\sinh \gamma l}} \quad \text{--- (5)}$$

so that $\frac{Z_1}{2Z_2} + 1 = \cosh \gamma l$

(15)

$$\Rightarrow \frac{1}{Z_2} \left(\frac{Z_1}{2} + Z_2 \right) = \cosh \gamma l$$

$$\Rightarrow \frac{Z_1}{2} + Z_2 = Z_2 \cosh \gamma l$$

$$\Rightarrow \frac{Z_1}{2} = Z_2 \cosh \gamma l - Z_2 = Z_2 (\cosh \gamma l - 1)$$

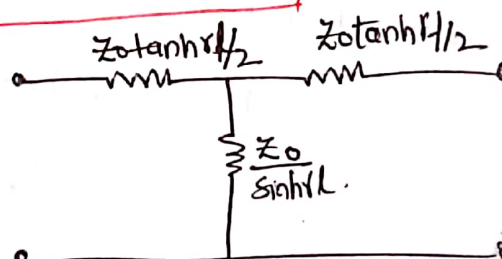
$$\Rightarrow \frac{Z_1}{2} = \frac{Z_0}{\sinh \gamma l} (\cosh \gamma l - 1) \quad \text{from eq (5)}$$

$$= \frac{Z_0 (2 \cosh^2 \gamma l / 2 + 1 - 1)}{2 \cosh \gamma l / 2 \sinh \gamma l / 2}$$

$$= \frac{Z_0 2 \cosh^2 \gamma l / 2}{2 \cosh \gamma l / 2 \sinh \gamma l / 2}$$

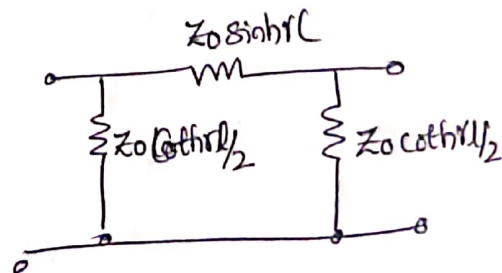
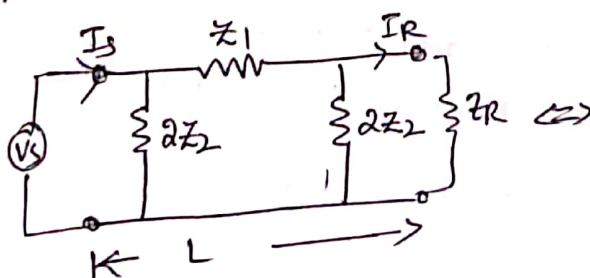
$$= Z_0 \frac{\cosh \gamma l / 2}{\sinh \gamma l / 2} = Z_0 \tanh \gamma l / 2$$

$$\therefore \frac{Z_1}{2} = Z_0 \tanh \gamma l / 2 \quad \text{--- (6)}$$



fig(c) Equivalent T-network of T/L.

Hence similarly, given Z_0 and γ of a T/L, its equivalent π -network of a T/L is obtained as



fig(d) Equivalent π -network of a T/L.

1.8 Infinite Transmission line

A transmission line, whose length is infinite is called infinite transmission line.

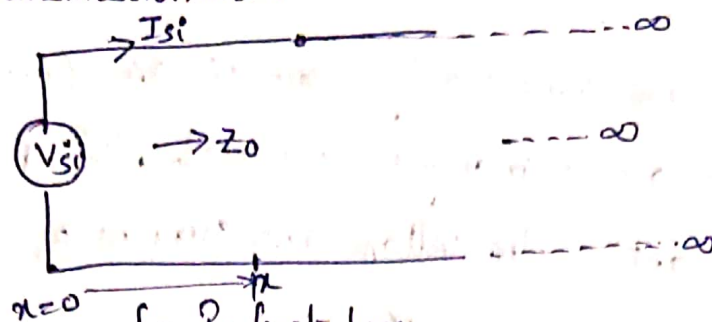


Fig Infinite line.

When AC voltage is applied to the sending end of an infinite line, a finite current will flow due to the capacitance C and the leakage conductance G between the two wires of the line. This current will not reach the other end due to infinite attenuation.

Let us consider a hypothetical transmission line as shown in fig:

Current at any point distance x from the sending end is given by $I = ce^{rx} + de^{-rx}$ — (1)

at $x=0$, $I = I_s$

$$I_s = c + d \quad \text{--- (2)}$$

at $x = \infty$, $I = 0$

$$0 = c \times \infty + d \times 0$$

$$\therefore c = 0$$

$$\left(\because e^{-\infty} = 0 \right. \\ \left. e^{\infty} = \infty \right)$$

$\therefore d = I_s$ — (3), Putting (2) & (3) in eq (1), we get

$$\therefore \boxed{I = I_s e^{-rx}} \quad \text{--- (4)}$$

eq (4) gives ~~current~~ current at any point of an infinite line.

Similarly, the voltage at any point of an infinite line can be (13)

obtained as

$$V = V_{si} e^{-\gamma x} \quad \text{--- (5)}$$

- (2) $\therefore$ The wave travels only in one direction (i.e. forward direction) and no standing waves exist. If γ is the propagation constant, the voltage and current equations of the infinite lines are

$$\begin{aligned} V &= V_{si} e^{-\gamma x} \\ I &= I_{si} e^{-\gamma x} \end{aligned}$$

- (3) The ratio of voltage applied to the current flowing will give the input impedance of an Infinite line. This input impedance is known as characteristic impedance of a line and is denoted by Z_0 .

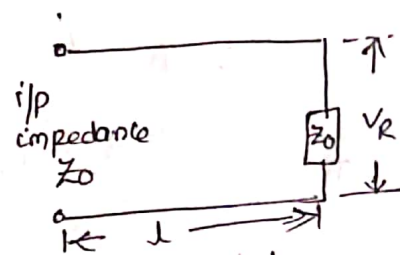
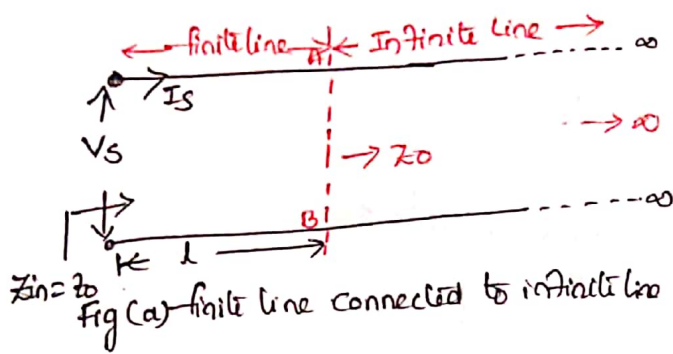
$$\therefore Z_0 = \frac{V_{si}}{I_{si}} \quad \text{--- (6)}$$

Where V_{si} and I_{si} are sending voltages and sending currents, respectively of an infinite line as shown in fig.

→

1.9 Infinite line is equivalent to a finite line terminated in its characteristic impedance Z_0

If a finite line is terminated by its characteristic impedance, the finite line will be equivalent to an infinite line as shown in fig(b). If a transmission of finite length is connected to an infinite line, these two lines together will behave as an infinite line as shown in fig(a).



Let the length of the finite line as shown in fig(b) be 'l' and terminated by its characteristic impedance Z_0 (i.e. $Z_L = Z_0$) having voltage & current V_R and I_R respectively at the terminated end.

$$\therefore Z_0 = \frac{V_R}{I_R} \quad \text{--- (1)}$$

Putting $x=l$, $V=V_R$ and $I=I_R$ in transmission line equation, we get

$$V_R = V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l \quad \text{--- (2)}$$

$$I_R = I_s \cosh \gamma l - \frac{V_s}{Z_0} \sinh \gamma l \quad \text{--- (3)}$$

from (2) & (3)

$$\frac{V_R}{I_R} = \frac{V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l}{I_s \cosh \gamma l - \frac{V_s}{Z_0} \sinh \gamma l} = Z_0 \quad \text{from eq (1)}$$

$$\therefore Z_0 = \frac{V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l}{I_s \cosh \gamma l - \frac{V_s}{Z_0} \sinh \gamma l} \quad \text{--- (4)}$$

multiplying numerator and denominator by Z_0 on R.H.S. we get

$$Z_0 = \frac{Z_0 (V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l)}{Z_0 I_s \cosh \gamma l - V_s \sinh \gamma l}$$

$$(or) \quad I = \frac{V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l}{Z_0 I_s \cosh \gamma l - V_s \sinh \gamma l}$$

$$(or) \quad Z_0 I_s \cosh \gamma l - V_s \sinh \gamma l = V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l$$

$$\Rightarrow \quad Z_0 I_s (\cosh \gamma l + \sinh \gamma l) = V_s (\cosh \gamma l + \sinh \gamma l)$$

$$\Rightarrow \quad Z_0 I_s = V_s$$

$$\text{Therefore } \boxed{Z_0 = \frac{V_s}{I_s}} \quad \text{--- (5)}$$

But $\frac{V_s}{I_s}$ is the input impedance of the line

$$\therefore \boxed{Z_{in} = Z_0} \quad \text{--- (6)}$$

- * Thus the input impedance of a finite line terminated in its characteristic impedance Z_0 is the characteristic impedance of line
- * By def. the i/p impedance of an infinite line is the characteristic impedance of the line.
- * Therefore a finite line terminated in its Z_0 , is equivalent to an infinite line as both will have an i/p impedance of Z_0

————— x —————

1.10 Expression for Characteristic Impedance Z_0 of a line

(a)

Def. 1 The ratio of the voltage applied to the current flowing through the infinite line is called characteristic impedance of the line and it is denoted by Z_0 . It is measured in ohm.

$$\therefore \boxed{Z_0 = \frac{V_{si}}{I_{si}}} \quad \text{--- (1)}$$

To obtain Z_0 in terms of Primary constants R, L, C & G consider the equation

$$- \frac{dV}{dx} = (R + j\omega L) I \quad \text{--- (2)}$$

However, the voltage and current eqns at any point of an infinite line are given by

$$V = V_{si} e^{-\gamma x} \quad \text{--- (3)}$$

$$I = I_{si} e^{-\gamma x} \quad \text{--- (4)}$$

Substitute (3) & (4) in eq (2) we get

$$- \frac{d}{dx} [V_{si} e^{-\gamma x}] = (R + j\omega L) I_{si} e^{-\gamma x}$$

$$\Rightarrow - (-\gamma V_{si} e^{-\gamma x}) = (R + j\omega L) I_{si} e^{-\gamma x}$$

$$\Rightarrow + V_{si} e^{-\gamma x} = \frac{(R + j\omega L) I_{si} e^{-\gamma x}}{\gamma}$$

$$\Rightarrow \frac{V_{si}}{I_{si}} = \frac{R + j\omega L}{\gamma}$$

$$\because \gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$\Rightarrow \frac{V_{si}}{I_{si}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \quad \text{--- (5)}$$

By def. input impedance Z_s of a infinite line gives the characteristic impedance Z_0 , i.e

$$Z_0 = \frac{V_{si}}{I_{si}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \quad \text{or} \quad \boxed{Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{Z}{Y}}} \quad \text{--- (6)}$$

(21)

The characteristic impedance is found physically by the size and spacing of the conductors and by dielectric constant of the insulator employed between the conductors.

∴ i) for parallel wire lines

$$Z_0 = \frac{276}{\sqrt{\epsilon_r}} \log_{10} \frac{S}{r} \Omega$$

ii, for coaxial cable

$$Z_0 = \frac{138}{\sqrt{\epsilon_r}} \log_{10} \frac{D}{d} \Omega$$

where ϵ_r = dielectric constant of the insulation

D = inner diameter of outer conductor of the co-axial cable

d = diameter of inner ~~radius~~ conductor

S = centre to centre spacing between two conductors.

r = radius of the either conductor.

— x —

1.10(b)
Propagation constant, γ :

Def:- The propagation constant of a uniform transmission line per unit length is defined as the natural logarithm of the vector ratio of the steady state voltages or currents entering and leaving the structure i.e

i) $\gamma = \ln\left(\frac{V_s}{V_R}\right) = \ln\left(\frac{I_s}{I_R}\right) - \text{① nepers}$

In decibels,

ii, $\gamma = 20 \log_{10}\left(\frac{V_s}{V_R}\right) = 20 \log_{10}\left(\frac{I_s}{I_R}\right) - \text{② decibels}$

ii)

$$\boxed{\gamma = \alpha + j\beta = \sqrt{(R+j\omega L)(G+j\omega C)}} \text{ in terms of primary ccts.} \quad (3)$$

where α = attenuation constant, dB/m
 β = phase shift constant, rad/m.

Note: $\boxed{1 \text{ nepel} = 8.686 \text{ dB}, 1 \text{ rad} = 57.3^\circ}$

1.10C) Derivation for Attenuation constant (α) and phase constant (β).

Consider propagation constant $\gamma = \alpha + j\beta = \sqrt{(R+j\omega L)(G+j\omega C)}$ — (1)

Squaring on both sides of equation (1), we get

$$\gamma^2 = (\alpha + j\beta)^2 = (\sqrt{(R+j\omega L)(G+j\omega C)})^2$$

$$\Rightarrow \alpha^2 - \beta^2 + 2j\beta\alpha = (R+j\omega L)(G+j\omega C)$$

$$\Rightarrow \alpha^2 - \beta^2 + 2j\beta\alpha = RG - \omega^2 LC + j(\omega RC + G\omega L) \quad (2)$$

Equating real part & imaginary part of eq (2)

$$\alpha^2 - \beta^2 = RG - \omega^2 LC \quad (3)$$

$$2\alpha\beta = \omega RC + G\omega L \quad (4)$$

we know that the square magnitude of γ is

$$|\gamma|^2 = |\alpha + j\beta|^2 = \alpha^2 + \beta^2 = \sqrt{(R^2 + \omega^2 L^2)(G^2 + \omega^2 C^2)} \quad (5)$$

Solving equation (3) and (5), we have α and β values. [Note: Add eq (3) & (5) to get ' α ' value]

$$\text{i.e. } 2\alpha^2 = (RG - \omega^2 LC) + \sqrt{(R^2 + \omega^2 L^2)(G^2 + \omega^2 C^2)}$$

$$\Rightarrow \alpha^2 = \frac{1}{2} [RG - \omega^2 LC + \sqrt{(R^2 + \omega^2 L^2)(G^2 + \omega^2 C^2)}]$$

then.

$$\Rightarrow \boxed{\alpha = \sqrt{\frac{1}{2} [RG - \omega^2 LC + \sqrt{(R^2 + \omega^2 L^2)(G^2 + \omega^2 C^2)}]}} \quad (6)$$

Attenuation
Const.

ly from eq (3) & (4)

(note & substitute Eq (3) and (4) to get β value)

(23)

$$-2\beta^2 = (RG - \omega^2 LC) - \sqrt{(R^2 + \omega^2 L^2)(G^2 + \omega^2 C^2)}$$

$$\Rightarrow \beta^2 = \frac{1}{2} [(\omega^2 LC - RG) + \sqrt{(R^2 + \omega^2 L^2)(G^2 + \omega^2 C^2)}]$$

then $\Rightarrow \beta = \sqrt{\frac{1}{2} [(\omega^2 LC - RG) + \sqrt{(R^2 + \omega^2 L^2)(G^2 + \omega^2 C^2)}]}$ (7)

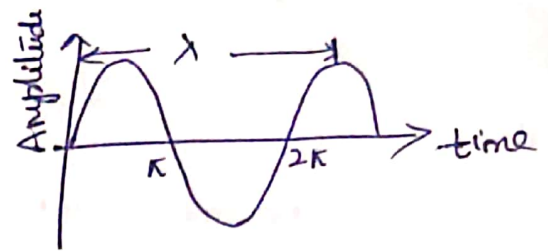
- phase constant of line

1.11 Other units of Transmission lines:

1. Wavelength (λ): is defined as the distance that a wave travels along the line in order that the total phase shift is 2π radians.

By def: $\beta\lambda = 2\pi$

$$\lambda = \frac{2\pi}{\beta}, \text{ metre (m)}$$



where β is the phase shift

2. Phase velocity or velocity of propagation: v_p :-

It is defined as the velocity with which a signal of single frequency propagates along the line. It is denoted by v_p , its units are km/sec.

$$\lambda = v_p t = \frac{v_p}{f} \quad (\because t = 1/f)$$

$$\Rightarrow v_p = f\lambda$$

$$\Rightarrow v_p = \frac{2\pi}{\beta} f = \frac{\omega}{\beta} \quad \left(\because \lambda = \frac{2\pi}{\beta} \right)$$

$$(\omega = 2\pi f)$$

$$\therefore v_p = \frac{\omega}{\beta}$$

3. Group velocity v_g : The velocity at which the envelope of the complex wave propagates along a line is called group velocity. It is denoted by v_g . It is defined as the ratio of the change in angular frequency to the change in phase constant of the wave.

$$\therefore v_g = \frac{\omega_2 - \omega_1}{\beta_1 - \beta_2} = \frac{d\omega}{d\beta} \quad \text{m/s}$$

where $\frac{d\omega}{d\beta}$ is the cut-off freq.

or group velocity v_g is always less than phase velocity v_p .

1.12 Relationship between Group velocity v_g and phase velocity v_p . (25)

We know that when a wave travels along a line, the phase velocity is

$$v_p = \frac{\omega}{\beta} \quad \text{--- (1)}$$

Differentiating w.r. to ω , we get

$$\frac{dv_p}{d\omega} = \frac{1 \cdot \beta - \omega \frac{d\beta}{d\omega}}{\beta^2} = \frac{1}{\beta} - \frac{\omega}{\beta^2} \frac{d\beta}{d\omega}$$

$$\Rightarrow \quad = \frac{1}{\beta} - \frac{1}{\beta} \left(\frac{\omega}{\beta} \right) \frac{d\beta}{d\omega} \quad \because v_p = \frac{\omega}{\beta}$$

$$\Rightarrow \quad \frac{dv_p}{d\omega} = \frac{1}{\beta} - \frac{1}{\beta} v_p \left(\frac{d\beta}{d\omega} \right)$$

$$\Rightarrow \quad \beta \frac{dv_p}{d\omega} = 1 - \frac{v_p}{v_g} \quad \because v_g = \frac{d\omega}{d\beta}$$

$$\Rightarrow \quad \frac{v_p}{v_g} = 1 - \beta \frac{dv_p}{d\omega}$$

$$\Rightarrow \quad v_g = \frac{v_p}{1 - \beta \frac{dv_p}{d\omega}} \quad \text{--- (2)}$$

Substituting the value of β , we get

$$= \boxed{v_g = \frac{v_p}{1 - \frac{\omega}{v_p} \left(\frac{dv_p}{d\omega} \right)}} \quad \text{--- (3)}$$

If v_p is constant w.r. to ω , then

$$\frac{dv_p}{d\omega} = 0 \Rightarrow \boxed{v_g = v_p}$$

Thus if v_p is independent of frequency, it becomes group velocity v_g

1.1/3 Lossless Transmission line ($R=0, G=0$)

A transmission line is said to be lossless if the conductors of the line are perfect ($\sigma_c = \infty$) and the dielectric medium between the line is lossless ($\sigma_d = 0$).

Also a line is said to be lossless, if

$$R = G = 0$$

Conditions for lossless line

1. Propagation constant $\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$

for lossless, $\alpha = 0$

$\therefore \boxed{\gamma = j\beta}$ purely imaginary.
-①

Also, we have

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} \quad \because R = G = 0$$
$$= \sqrt{(j\omega L)(j\omega C)} \quad [\because \gamma = j\beta = j\omega \sqrt{LC}]$$

$\boxed{\gamma = j\omega \sqrt{LC}}$ -②

2. phase shift constant $\boxed{\beta = \omega \sqrt{LC}}$ -③

3. Characteristic impedance

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{j\omega L}{j\omega C}} = \sqrt{\frac{L}{C}} \quad -④$$

4. phase velocity $v_p = \frac{\omega}{\beta} = \frac{\omega}{\omega \sqrt{LC}} = \frac{1}{\sqrt{LC}} \quad -⑤$

Summary: $\boxed{\gamma = j\omega \sqrt{LC}, \alpha = 0, \beta = \omega \sqrt{LC}, Z_0 = \sqrt{\frac{L}{C}}, v_p = \frac{1}{\sqrt{LC}}}$

Thus Z_0 and v_p depends only on L and C values.

1.14 Distortionless line:

A transmission line is said to be distortionless when the attenuation constant (α) is independent of frequency and phase constant, β is linearly dependent on the frequency.

$$\boxed{\frac{R}{L} = \frac{G}{C}}$$

we have

$$\gamma = \sqrt{(R+j\omega L)(G+j\omega C)} = \alpha + j\beta \quad \text{--- (1)}$$

$$= \sqrt{L\left(\frac{R}{L} + j\omega\right)C\left(\frac{G}{C} + j\omega\right)}$$

$$= \sqrt{LC} \sqrt{\left(\frac{R}{L} + j\omega\right)\left(\frac{G}{C} + j\omega\right)} \quad \text{--- (2)}$$

To make α in the above equation independent of frequency,

$$\text{let } \frac{R}{L} = \frac{G}{C} \quad \text{--- (3)}$$

$$\text{Then } \gamma = \alpha + j\beta = \sqrt{LC} \sqrt{\left(\frac{R}{L} + j\omega\right)^2} = \sqrt{LC} \left(\sqrt{\frac{R}{C} + j\omega}\right)^2$$

$$\Rightarrow \alpha + j\beta = \sqrt{LC} \left(\frac{R}{L} + j\omega\right) = \sqrt{LC} \left(\frac{G}{C} + j\omega\right) \quad \text{--- (4)}$$

Equating real part and imaginary part

$$\alpha = \sqrt{LC} \frac{R}{L} \text{ (or) } \sqrt{LC} \frac{G}{C}$$

$$\boxed{\alpha = R \sqrt{\frac{C}{L}} \text{ or } G \sqrt{\frac{L}{C}}} \quad \text{--- (5)}$$

$$\boxed{\beta = \omega \sqrt{LC}} \quad \text{--- (6)}$$

Also

characteristic impedance $Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$

(28)

$$\Rightarrow Z_0 = \sqrt{\frac{L(R/L + j\omega)}{C(G/C + j\omega)}}$$

$$\therefore \frac{R}{L} = \frac{G}{C}$$

$$\therefore \boxed{Z_0 = \sqrt{\frac{L}{C}}} \quad \text{--- (7)}$$

phase velocity $v_p = \frac{\omega}{\beta} = \frac{\omega}{\omega \sqrt{LC}}$

$$\therefore \boxed{v_p = \frac{1}{\sqrt{LC}}} \quad \text{--- (8)}$$

Therefore the condition for distortionless line is

$$\boxed{\frac{R}{L} = \frac{G}{C}} \quad \text{or } RC = GL$$

- (2) In distortionless line, the α , Z_0 , v_p are independent of freq. and β varies linearly with frequency.

1.15 High frequency T/L ($\omega L \gg R$ and $\omega C \gg G$) or RF lines (29)

The propagation constant of a uniform T/L is given by

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \alpha + j\beta \quad \text{--- (1)}$$

If frequency is very high, then $\omega R \gg G$ and $\omega L \gg R$, then

$$\begin{aligned} \gamma &= \sqrt{j\omega L \left(1 + \frac{R}{j\omega L}\right) j\omega C \left(1 + \frac{G}{j\omega C}\right)} \\ &= j\omega \sqrt{LC} \left(1 + \frac{R}{j\omega L}\right)^{1/2} \left(1 + \frac{G}{j\omega C}\right)^{1/2} \quad \text{--- (2)} \end{aligned}$$

By using binomial theorem, eq (2) can be expanded as

$$\gamma = j\omega \sqrt{LC} \left(1 + \frac{R}{j2\omega L} + \frac{1}{2} \left(\frac{R}{j\omega L}\right)^2 + \dots\right) \left(1 + \frac{G}{j2\omega C} + \frac{1}{2} \left(\frac{G}{j\omega C}\right)^2 + \dots\right)$$

$$\Rightarrow \gamma = j\omega \sqrt{LC} \left(1 + \frac{R}{j2\omega L} + \frac{G}{j2\omega C} + \frac{RG}{j4\omega^2 LC} + \dots\right)$$

neglecting higher order terms of ω

$$\Rightarrow \gamma = j\omega \sqrt{LC} \left(1 + \frac{R}{j2\omega L} + \frac{G}{j2\omega C}\right) \quad \text{--- (3)}$$

$$\Rightarrow \gamma = j\omega \sqrt{LC} + j\omega \sqrt{LC} \cdot \frac{R}{j2\omega L} + j\omega \sqrt{LC} \cdot \frac{G}{j2\omega C}$$

$$\Rightarrow \gamma = j\omega \sqrt{LC} + \frac{R}{2} \sqrt{\frac{C}{L}} + \frac{G}{2} \sqrt{\frac{L}{C}} \quad \text{--- (4)}$$

We know $\gamma = \alpha + j\beta$.

$\therefore$ Equating real part and imaginary part

$$\boxed{\alpha = \frac{R}{2} \sqrt{\frac{C}{L}} + \frac{G}{2} \sqrt{\frac{L}{C}}} \quad \text{--- (5)}$$

$$\boxed{\beta = \omega \sqrt{LC}} \quad \text{--- (6)}$$

$$v_p = \frac{\omega}{\beta} = \frac{\omega}{\omega \sqrt{LC}} = \frac{1}{\sqrt{LC}}$$

(30)

$$\therefore \boxed{v_p = \frac{1}{\sqrt{LC}}} \quad \text{m/s} \quad \text{--- (7)}$$

The characteristic impedance Z_0 is given by

$$Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$$

Since $\omega L \gg R$, $\omega C \gg G$,

$$Z_0 \approx \sqrt{\frac{j\omega L}{j\omega C}} = \sqrt{\frac{L}{C}}$$

$$\therefore \boxed{Z_0 = \sqrt{\frac{L}{C}}} \quad \Omega \quad \text{--- (8)}$$

④ Hence α , Z_0 , v_p are independent of frequency and β varies linearly with frequency. Hence high frequency or RF lines are distortionless. $\rightarrow$

1.16 Telephone Cables ($\omega L \ll R$ and $G \ll \omega C$)

A telephone cable has insulated wires, twisted in pairs. It has negligible values of inductance L and conductance G , over the range of frequencies. i.e. $\omega L \ll R$ and $G \ll \omega C$.

$$\text{The propagation constant } \gamma = \alpha + j\beta = \sqrt{(R+j\omega L)(G+j\omega C)} \quad \text{--- (1)}$$

Since L and G are negligible, eq (1) becomes

$$\gamma \approx \sqrt{R(j\omega C)} = \sqrt{j\omega RC} = \sqrt{\omega RC} \angle 90^\circ = \sqrt{\omega RC} \angle 45^\circ$$

$$= \sqrt{\omega RC} (\cos 45^\circ + j \sin 45^\circ)$$

$$= \sqrt{\omega RC} \left(\frac{1}{\sqrt{2}} + j \frac{1}{\sqrt{2}} \right)$$

$$\text{Hence } \gamma = \alpha + j\beta = \sqrt{\frac{\omega RC}{2}} (1 + j) \quad \text{--- (1)}$$

Equating real part & img. part yields

$$\alpha = \sqrt{\frac{\omega RC}{2}} \quad \text{--- (2)}$$

$$\beta = \sqrt{\frac{\omega RC}{2}} \quad \text{--- (3)}$$

$$v_p = \frac{\omega}{\beta} = \frac{\omega}{\sqrt{\frac{\omega RC}{2}}} = \sqrt{\frac{2\omega}{RC}} \quad \text{--- (4)}$$

Characteristic impedance Z_0 will be

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \approx \sqrt{\frac{R}{j\omega C}} = \sqrt{\frac{-jR}{\omega C}} = \sqrt{\frac{R}{\omega C}} \angle -90^\circ$$

$$\therefore Z_0 = \sqrt{\frac{R}{\omega C}} \angle -45^\circ \quad \text{--- (5)}$$

From eq (2) and (4), it is observed that both α and ν_p are dependent on frequency. Thus the higher frequencies are attenuated and travel faster than the lower frequencies resulting considerable frequency and delay distortion.

— x —

UHF Lines as circuit elements:-

→ At ultra high frequency, the transmission line becomes lossless. The short length SC and O.C transmission lines can be used as circuit elements.

SC (short ckt) lines at different lengths:-

for lengths $(0 < l < \frac{\lambda}{4})$

We know that $Z_{sc} = jZ_0 \tan \beta l$. ($\because$ lossless)

Z_{sc} is purely reactive. So if L_{eq} = equivalent Inductance then

$$j\omega L_{eq} = jZ_0 \tan \beta l$$

$$\therefore \boxed{L_{eq} = \frac{Z_0}{\omega} \tan \beta l}$$

therefore the line acts as an Inductor.



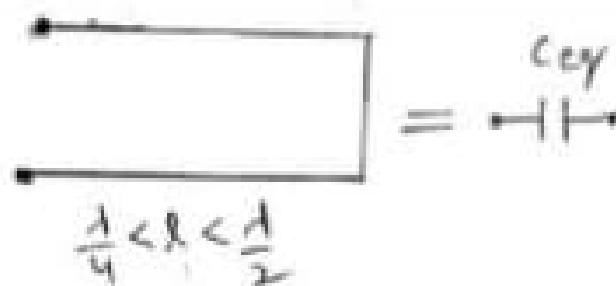
for lengths $(\frac{\lambda}{4} < l < \frac{\lambda}{2})$

$$\frac{1}{j\omega C_{eq}} = jZ_0 \tan \beta l$$

$$\Rightarrow C_{eq} = \frac{1}{j\omega Z_0 \tan \beta l}$$

$$\therefore \boxed{C_{eq} = \frac{-1}{\omega Z_0 \tan \beta l}}$$

the line acts as capacitor.



c) for $l = \frac{\lambda}{4}$:- We know that $Z_{sc} = jZ_0 \tan \beta l$.

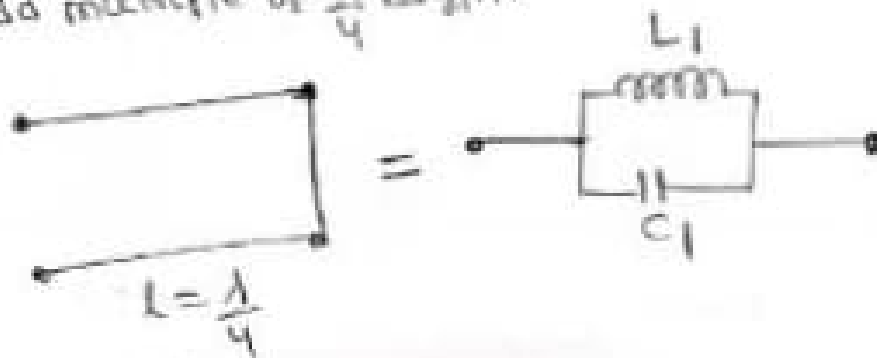
$$\beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$$

$$(\because \tan \frac{\pi}{2} = \infty)$$

$$\Rightarrow Z_{sc} = jZ_0 \tan \frac{\pi}{2}$$

$$\therefore \boxed{Z_{sc} = \infty}$$

- The SC line has infinite impedance (input impedance)
- It acts as a parallel (or) anti resonance circuit at every odd multiple of $\frac{\lambda}{4}$ length.



d) for $l = \frac{\lambda}{2}$:-

$$Z_{sc} = jZ_0 \tan \beta l, \quad \beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$$

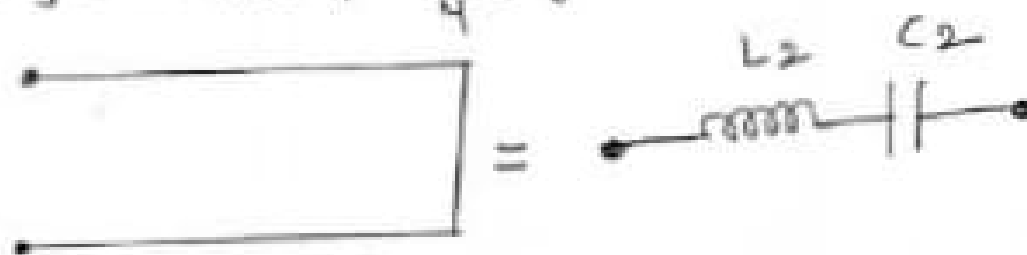
$$Z_{sc} = jZ_0 \tan \pi$$

$$(\because \tan \pi = 0)$$

$$= 0$$

$$\therefore \boxed{Z_{sc} = 0}$$

- therefore for $l = \frac{\lambda}{2}$ The SC line has zero input impedance. It acts as a series resonant CKT at every even multiples $\frac{\lambda}{4}$ length.



✓ OC (open ckt) lines at different lengths:-

① for lengths $(0 < l < \frac{\lambda}{4})$:- Z_{oc} is also purely reactive.

We know that,

$$Z_{oc} = -j Z_0 \cot \beta l.$$

If C_{eq} = Equivalent capacitance then

$$\frac{1}{j\omega C_{eq}} = -j Z_0 \cot \beta l.$$

$$\therefore C_{eq} = \frac{1}{-j^2 \omega Z_0 \cot \beta l}$$

$$\therefore \boxed{C_{eq} = \frac{1}{\omega Z_0 \cot \beta l}}$$

The line acts as capacitor

$$(\because j^2 = -1)$$

$$(-j^2 = 1)$$

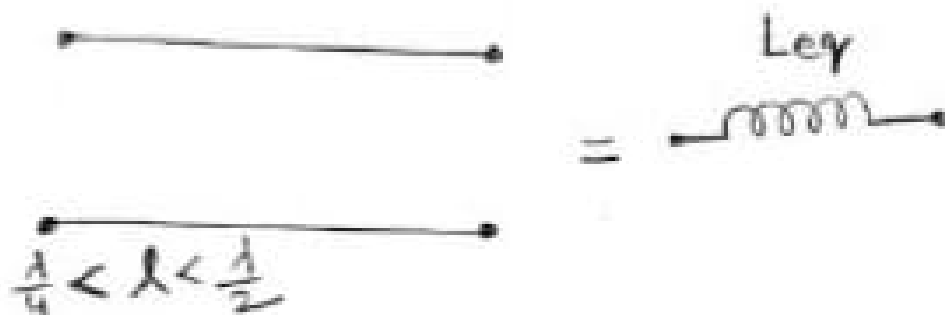


② for length $(\frac{\lambda}{4} < l < \frac{\lambda}{2})$ We know that $Z_{oc} = -j Z_0 \cot \beta l$

$$j\omega L_{eq} = -j Z_0 \cot \beta l.$$

$$\boxed{L_{eq} = -\frac{Z_0}{\omega} \cot \beta l}$$

The line acts as an Inductor.



i) OC (open ckt) lines at different lengths:-

② for lengths $(0 < l < \frac{\pi}{4})$:- Z_{oc} is also purely reactive.
We know that,

$$Z_{oc} = -j Z_0 \cot \beta l.$$

If C_{eq} = Equivalent capacitance then

$$\frac{1}{j\omega C_2} = -jZ_2 \cot \beta l$$

$$\therefore C_{eq} = \frac{1}{-j\omega Z_0 \cot \beta l}$$

$$\therefore C_{ef} = \frac{1}{\omega Z_0 \cot \beta l}$$

The line acts as capacitor

$$(z^2 - 1)$$

$$(\zeta^2 = 1)$$

$$= \frac{C_{eq}}{2}$$

$$0 < \lambda < \frac{\lambda}{4}$$

(b) for length $(\frac{\lambda}{4} < l < \frac{\lambda}{2})$ We know that $Z_{oc} = -jZ_0 \cot \beta l$
 $j\omega L_{eq} = -jZ_0 \cot \beta l.$

$$\frac{1}{2} \omega_1 - \omega_2 = -\frac{1}{2} \omega_0 \cot \beta_1.$$

$$L_{eq} = -\frac{Z_0}{\omega} \cot \beta l$$

The line acts as an Inductor.

L_{eq}

$$\frac{1}{4} < 1 < \frac{1}{2}$$

③ for length $l = \frac{\lambda}{4}$:- $\beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$

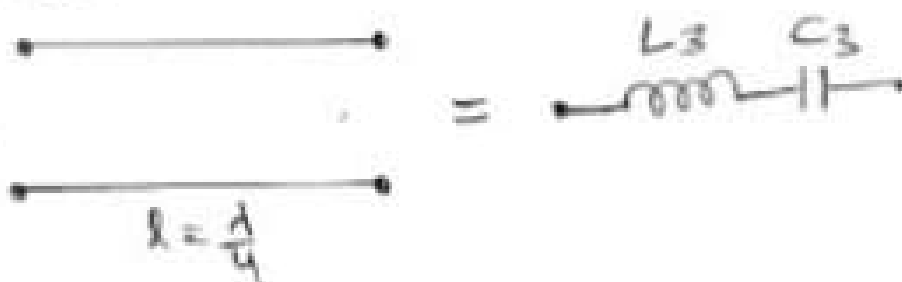
$$Z_{oc} = -jZ_0 \cot \beta l.$$

$$Z_{oc} = -jZ_0 \cot \frac{\pi}{2} = 0.$$

$$(\because \cot \frac{\pi}{2} = \frac{\cos 90^\circ}{\sin 90^\circ} = 0)$$

$$\boxed{Z_{oc} = 0}$$

→ The o.c line has zero input impedance. It acts as series resonant circuit at odd multiples of $\frac{\lambda}{4}$ length.



④ for length $l = \frac{\lambda}{2}$:-

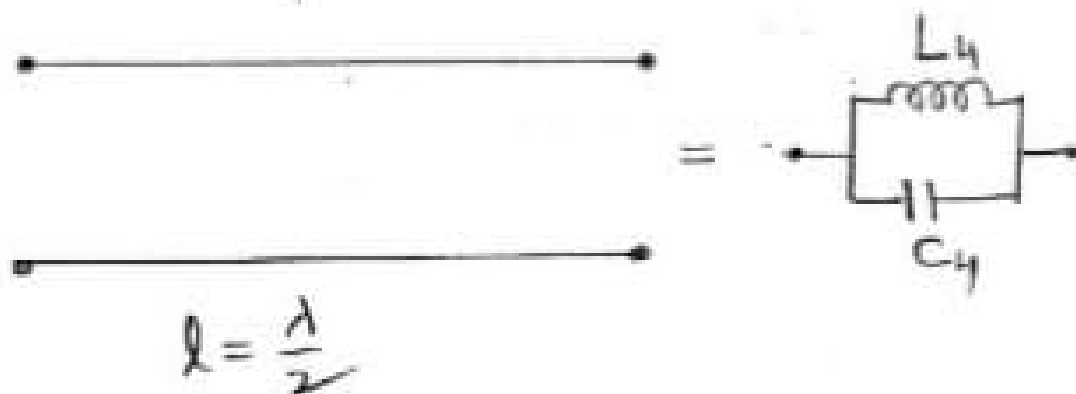
$$\beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$$

$$Z_{oc} = -jZ_0 \cot \beta l = -jZ_0 \cot \pi$$

$$(\because \cot \pi = \infty)$$

$$\boxed{Z_{oc} = \infty}$$

→ The o.c line has infinite input impedance. It acts as parallel resonance circuit at even multiples of $\frac{\lambda}{4}$ length



Impedance Transformations :-

- Input impedance of a transmission line depends on its length. Thus a short length transmission line is used as an "impedance matching device" (or) impedance transformation.
- If the load impedance is not equal to complex conjugate of source impedance, a short length transmission line is added to the line to give "maximum power transfer".

The important impedance transformations are

- (1) eight wave ($\frac{\lambda}{8}$) transmission line
- (2) The quarter wave ($\frac{\lambda}{4}$) transmission line
- (3) The half wave ($\frac{\lambda}{2}$) transmission line.

D) Eight Wave Transmission line ($\frac{\lambda}{8}$) :-

For eight wave length $l = \frac{\lambda}{8}$

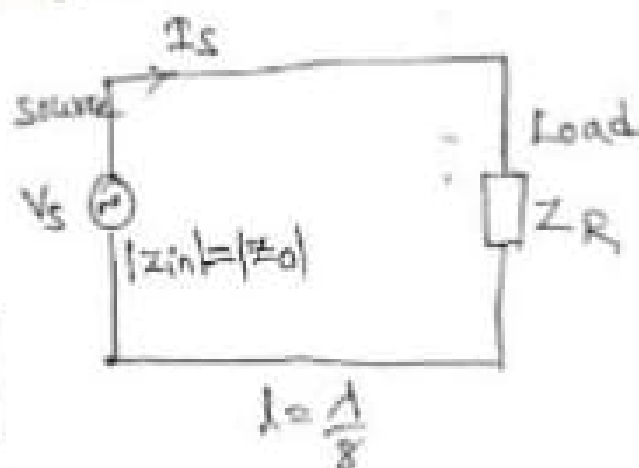
λ = Wavelength,

We know that input impedance of transmission line is

$$Z_{in} = Z_0 \left[\frac{Z_R + Z_0 \tanh \Gamma l}{Z_0 + Z_R \tanh \Gamma l} \right]$$

For lossless $\alpha = 0$, and $\Gamma = j\beta$.

$$Z_{in} = Z_0 \left[\frac{Z_R + j Z_0 \tan \beta l}{Z_0 + j Z_R \tan \beta l} \right]$$



For length $l = \frac{\lambda}{4}$

$$(\because \tan \frac{\pi}{4} = 1)$$

$$\Rightarrow \beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{4}$$

$$\therefore Z_{in} = Z_0 \left[\frac{Z_R + j Z_0 \tan \frac{\pi}{4}}{Z_0 + j Z_R \tan \frac{\pi}{4}} \right] = Z_0 \left[\frac{Z_R + j Z_0}{Z_0 + j Z_R} \right]$$

$$|Z_{in}| = |Z_0| \left| \frac{Z_R + j Z_0}{Z_0 + j Z_R} \right|$$

$$\Rightarrow |Z_{in}| = |Z_0| \times \frac{\sqrt{Z_R^2 + Z_0^2}}{\sqrt{Z_0^2 + Z_R^2}}$$

$$\therefore |Z_{in}| = |Z_0|$$

The magnitude of i/p impedance is equal to magnitude of characteristic impedance.

Quarter Wave Transmission Line:- $\left(\frac{\lambda}{4}\right)$

For quarter wave $l = \frac{\lambda}{4}$

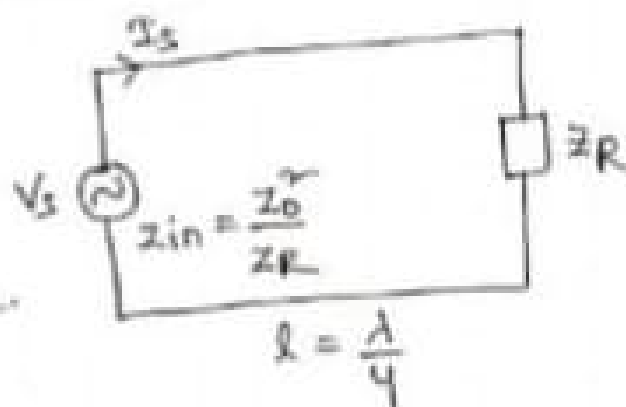
Consider a $\frac{\lambda}{4}$ transmission line terminated with Z_R .

We know that, for lossless line.

$$Z_{in} = Z_0 \left[\frac{Z_R + j Z_0 \tan \beta l}{Z_0 + j Z_R \tan \beta l} \right]$$

$$\text{for } l = \frac{\lambda}{4}, \beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{4} = \frac{\pi}{2}$$

$$Z_{in} = Z_0 \left[\frac{\frac{Z_R}{\tan \frac{\pi}{2}} + j Z_0}{\frac{Z_0}{\tan \frac{\pi}{2}} + j Z_R} \right] = Z_0 \left[\frac{\frac{Z_R}{\infty} + j Z_0}{\frac{Z_0}{\infty} + j Z_R} \right]$$



$$\therefore Z_{in} = Z_0 \left[\frac{\frac{Z_R}{Z_0} + j}{-\frac{Z_0}{Z_R} + j} \right] = Z_0 \left[\frac{0 + j Z_0}{0 + j Z_R} \right]$$

$$\therefore Z_{in} = Z_0 \left[\frac{j Z_0}{j Z_R} \right]$$

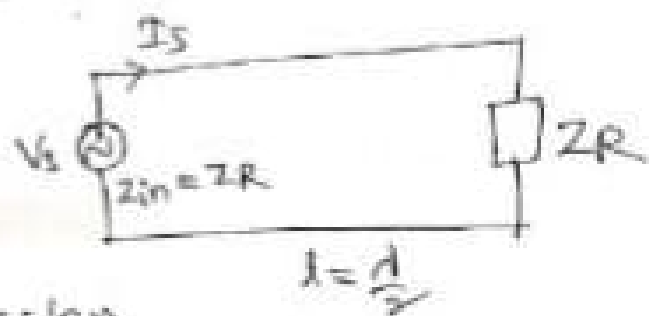
$$\boxed{Z_{in} = \frac{Z_0^2}{Z_R}} \quad (\text{or}) \quad \boxed{Z_0 = \sqrt{Z_{in} \cdot Z_R}}$$

Thus a $\frac{\lambda}{4}$ length line is considered as a transformer that matches a load of Z_R to a source impedance of Z_{in} .

2) Half Wave Transmission Line: ($\frac{\lambda}{2}$)

for half wave $l = \frac{\lambda}{2}$

$$\beta l = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$$



We know that,

Input Impedance of a lossless Transmission Line is

$$Z_{in} = Z_0 \left[\frac{Z_R + j Z_0 \tan \beta l}{-Z_0 + j Z_R \tan \beta l} \right] = Z_0 \left[\frac{Z_R + j Z_0 \tan \pi}{-Z_0 + j Z_R \tan \pi} \right]$$

$$\Rightarrow Z_{in} = Z_0 \left[\frac{Z_R + 0}{-Z_0 + 0} \right] = \cancel{Z_0} \times \frac{Z_R}{\cancel{Z_0}}$$

$$\therefore \boxed{Z_{in} = Z_R}$$

Thus The Input Impedance of a half Wave Line is equal to load impedance.

Stub matching :- "A stub is a piece of Transmission line. It can be short circuited or open circuited at the Load end (far end)".

- Stub has a pure reactance (or) susceptance. It is used to cancel out reactance (or) susceptance of a Transmission line.
- A Stub is used for Impedance matching.
- When a UHF line is terminated with a load impedance, which is not equal to the characteristic impedance, so mismatch occurs.
- Mismatch reduces efficiency and increases power loss.
- To avoid mismatching we have to add impedance matching devices between load and line.

Advantages :-

1. The length and characteristic impedance of line will be same.
2. Since the stub is added in shunt, there is no need to cut the line.
3. The susceptance (or) reactance of stub can be adjusted for "perfect matching".

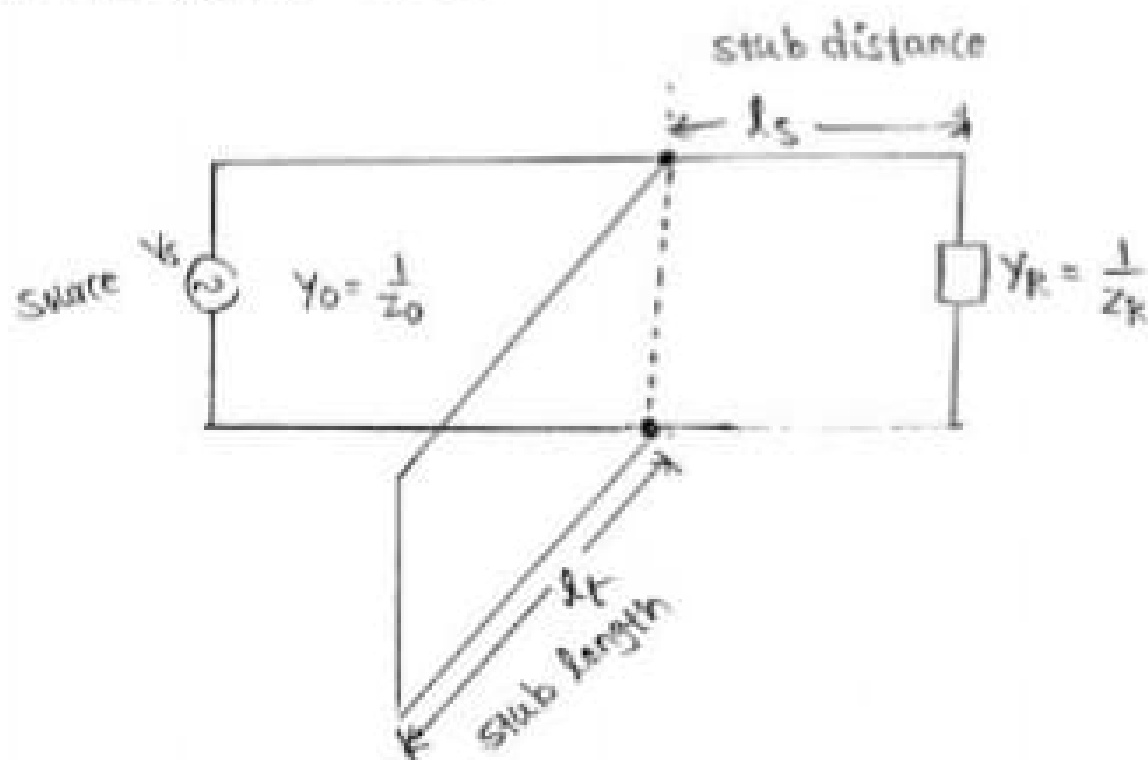
Methods of Stub matching :-

- 1) single stub matching
- 2) Double stub matching.

1) single stub matching :-

- In this method, to achieve impedance matching, an open or short circuited short length transmission line (stub) is connected in parallel to the main line at a certain distance from Load.
- Since the stub is connected in parallel, it is easy to use Admittance for analysis instead of Impedance.
- When Load admittance ($Y_R = \frac{1}{Z_R}$) is connected to the line and if it is not equal to characteristic admittance ($Y_0 = \frac{1}{Z_0}$) i.e. $Y_R \neq Y_0$ the mismatch occurs.

- So the standing waves exist on the line.
- The construction of single stub matching shown below



- When we move from load to source, the admittance on line varies from max to min, or min to max. Value depends on the length of line.
- At some point on the line, the real part of admittance (Y) equal to characteristic admittance (Y_0)

$$\text{Re}\{Y\} = \text{Re}\{Y_0\}.$$
- Generally a short ckt stub is preferred than open ckt stub, because
 - (i) short ckt stub provides strong and supports to construction and main line.
 - (ii) The short ckt stubs easily established, with metal plate
 - (iii) Radiation loss is very less compared to open ckt stub.

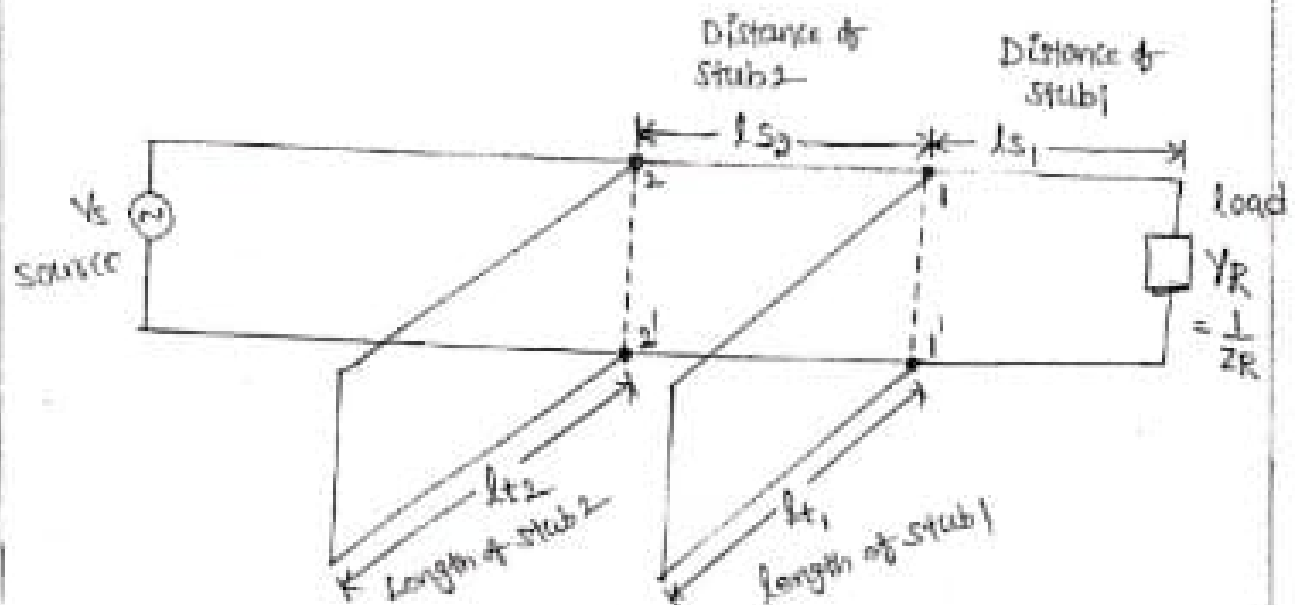
Disadvantages:-

1. The location and length of single stub matching depend on frequency, if the frequency & wave changes, the length and location of stub should be changed.
2. It is very difficult to place a stub on co-axial cables.

Double Stub matching :-

→ To overcome the disadvantages of single stub matching, two stubs can be used at different locations. This is called "Double Stub matching".

→ consider a double stub matching system consists of two short ckt stubs connected in parallel to line near the load.



→ The characteristic admittance of stubs should be equal to characteristic admittance of line (Y_0).

- In this system, the locations of stubs have to be chosen arbitrarily. But It is very difficult to design stub matching with arbitrarily locations. So the locations of stubs should be restricted. The admittance repeats at every $\frac{\lambda}{2}$ length.

The total distance never be more than (or) equal to $\frac{\lambda}{2}$.

$$\text{ie } l_{s1} + l_{s2} < \frac{\lambda}{2}$$

- So l_{s1} is in between 0.1λ and 0.5λ . Some times $l_{s1} = 0$ (load) can be chosen.

The space between two stubs is generally taken as $\frac{\lambda}{8}$, $\frac{\lambda}{4}$ (or) $\frac{3\lambda}{8}$ distance.

The total distance of the double stub matching should be small.

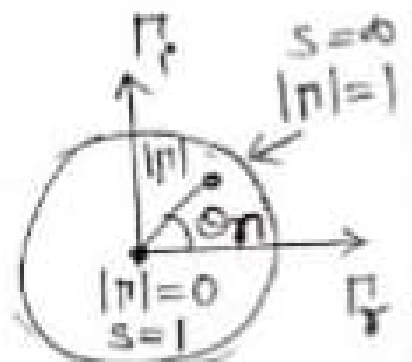
- ∴ The Double Stub matching consists of fixed stub locations. The impedance matching is done by finding the lengths of the stubs.

Smith chart:- phillip H. smith in the year 1939 developed a polar chart for calculating transmission line characteristics. this chart is called "smith chart".

Definition:- Smith chart is a polar plot of reflection coefficient in terms of normalised impedance $r + jx$.

(OR)
Smith chart is a graphical plot of normalised resistance and reactance in the reflection coefficient plane.

- It consists of two sets of orthogonal circles, which represents the values of normalised Impedance.
- one set of circles represents Resistive component 'r' called as "r-circles" and other set of circles represent reactive component 'x' called "x-circles".



construction:-

- The smith chart is constructed within a circle of unit radius ($|\Gamma| \leq 1$) as shown in the given figure.
- The construction of smith chart is based on the relation is

$$\Gamma = \frac{Z_R - Z_0}{Z_R + Z_0} \rightarrow \textcircled{1}$$

(OR)

$$\Gamma = |\Gamma| \angle \theta_\Gamma = \Gamma_r + j\Gamma_j \rightarrow \textcircled{2}$$

Where Γ_r and Γ_j are the real and imaginary parts of the reflection coefficient Γ .

For the Load Impedance Z_R , the normalized Load impedance is given by

$$z_n = Z_R = \frac{Z_R}{Z_0} = \gamma + jx \rightarrow (3) \quad \left(\because \gamma = \frac{R}{Z_0} \right. \\ \left. jx = \frac{jX}{Z_0} \right)$$

from eq. (1), (2)

$$\Gamma = \Gamma_V + j\Gamma_I = \frac{Z_R - Z_0}{Z_R + Z_0} \quad \left(\because Z_n = Z_R = \text{normalized Impedance} \right)$$

$$\Rightarrow \Gamma = \Gamma_V + j\Gamma_I = \frac{Z_0 \left(\frac{Z_R}{Z_0} - 1 \right)}{Z_0 \left(\frac{Z_R}{Z_0} + 1 \right)} = \frac{z_n - 1}{z_n + 1} \rightarrow (4)$$

$$\Rightarrow z_n - 1 = (\Gamma_V + j\Gamma_I)(z_n + 1)$$

$$\Rightarrow z_n - 1 = \Gamma_V z_n + \Gamma_V + j\Gamma_I z_n + j\Gamma_I$$

$$\Rightarrow z_n - \Gamma_V z_n - j\Gamma_I z_n = 1 + \Gamma_V + j\Gamma_I$$

$$\Rightarrow z_n (1 - \Gamma_V - j\Gamma_I) = 1 + \Gamma_V + j\Gamma_I$$

$$\therefore \boxed{z_n = \frac{(1 + \Gamma_V) + j\Gamma_I}{(1 - \Gamma_V) - j\Gamma_I}} \rightarrow (5)$$

Multiplying and dividing by $(1 - \Gamma_V) + j\Gamma_I$

$$\Rightarrow z_n = \gamma + jx = \frac{(1 + \Gamma_V) + j\Gamma_I}{(1 - \Gamma_V) - j\Gamma_I} \times \frac{(1 - \Gamma_V) + j\Gamma_I}{(1 - \Gamma_V) + j\Gamma_I} \quad (\because j^2 = -1)$$

$$\Rightarrow \gamma + jx = \frac{(1 - \Gamma_V^2) + j\Gamma_I(1 + \Gamma_V) + j\Gamma_I(1 - \Gamma_V) + j^2 \Gamma_I^2}{(1 - \Gamma_V)^2 - j^2 \Gamma_I^2}$$

$$\Rightarrow \gamma + jx = \frac{(1 - \Gamma_V^2) + j\Gamma_I + j\Gamma_I \cancel{\Gamma_V} + j\Gamma_I - j\Gamma_I \cancel{\Gamma_V} - \Gamma_I^2}{(1 - \Gamma_V)^2 + \Gamma_I^2}$$

$$\Rightarrow \gamma + j\chi = \frac{(1 - \Gamma_V^2 - \Gamma_I^2) + j2\Gamma_I}{(1 - \Gamma_V)^2 + \Gamma_I^2}$$

Normalizing and equating real and imaginary parts.
We obtain

$$\gamma = \frac{1 - \Gamma_V^2 - \Gamma_I^2}{(1 - \Gamma_V)^2 + \Gamma_I^2}, \quad \chi = \frac{2\Gamma_I}{(1 - \Gamma_V)^2 + \Gamma_I^2}$$

Rearranging above equations, then we get

$$\left[\Gamma_V - \frac{\gamma}{1+\gamma} \right]^2 + \Gamma_I^2 = \left[\frac{1}{1+\gamma} \right]^2 \rightarrow \textcircled{6}$$

r-circle

$$\left[\Gamma_V - 1 \right]^2 + \left[\Gamma_I - \frac{1}{\gamma} \right]^2 = \left[\frac{1}{\gamma} \right]^2 \rightarrow \textcircled{7}$$

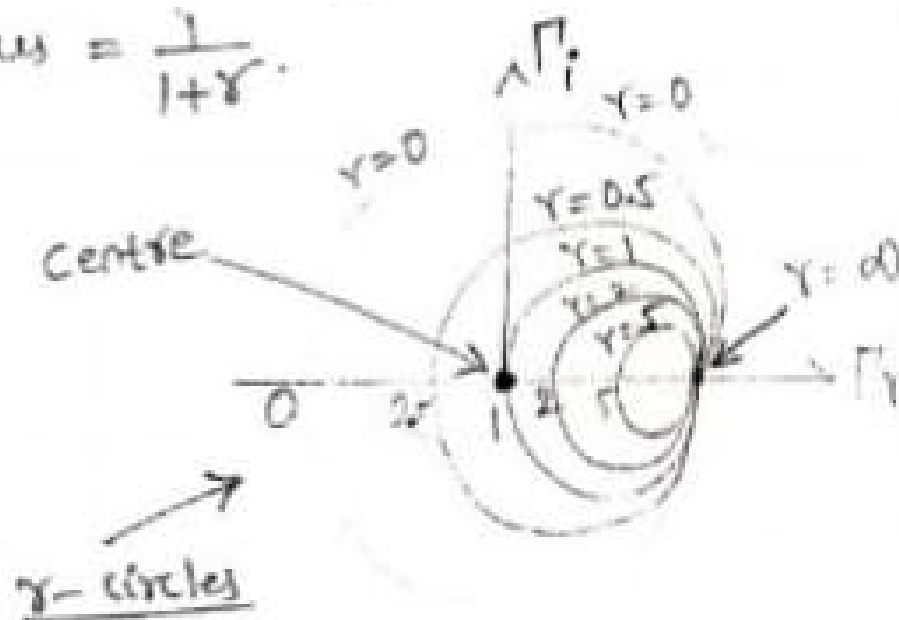
x-circle

The above equations are in the form of equation of circle. (ie) $(x-h)^2 + (y-k)^2 = a^2$ a = radius
(h,k) = center

$\therefore$ eq $\textcircled{6}$ is an 'r'-circle. (resistance circle)

Center at $(\Gamma_V, \Gamma_I) = \left(\frac{\gamma}{1+\gamma}, 0 \right)$

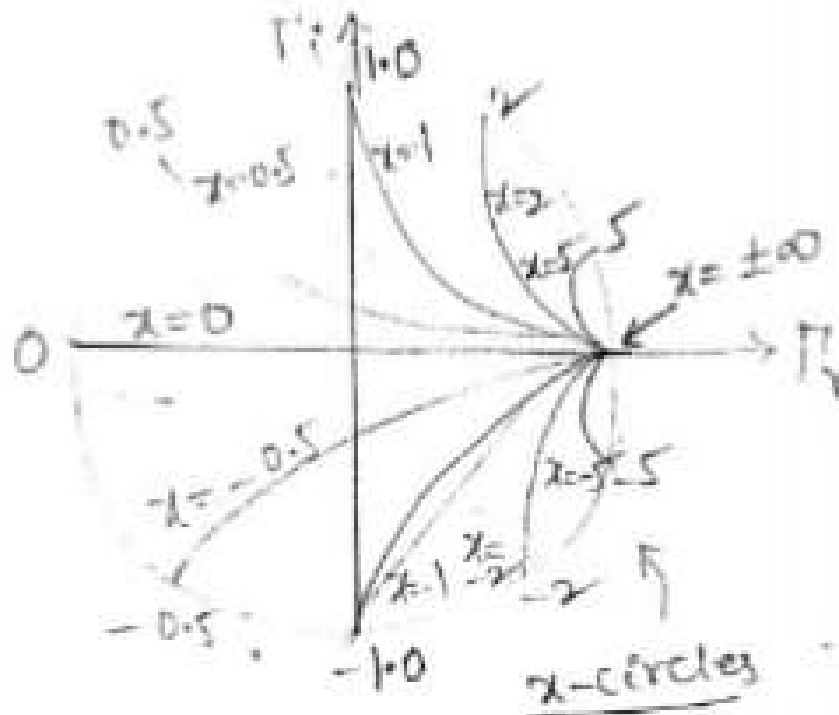
radius = $\frac{1}{1+\gamma}$.



Similarly $x_1 \oplus$ is an 'x'-circle (reactance circle).

Center at $(\Gamma_r, \Gamma_i) = (1, \frac{1}{x})$

radius = $\frac{1}{x}$



Applications of smith chart :-

The smith chart can be used to

1. Find the parameters of mismatched transmission lines.
2. Find the normalised impedance from normalized admittance
3. Find the normalized admittance from normalized Impedance.
4. Find Voltage Standing Wave Ratio (VSWR) for a given Load impedance.
5. Design stubs for Impedance matchings
6. Find the Reflection coefficient (Γ)

Reflection:- The phenomena of a wave reflected at load due to improper termination is called "Reflection".

* When $Z_R \neq Z_0$, the incident wave is reflected back at the load. So the power loss occurs due to reflection.

Reflection coefficient:- (Γ)

Reflection coefficient is defined as the ratio of the reflected voltage to the incident voltage. (OR) the ratio of reflected current to incident current.

→ It is a vector quantity.

$$\therefore \boxed{\Gamma = \frac{V_r}{V_i}} \quad (\text{OR}) \quad \boxed{\Gamma = -\frac{I_r}{I_i}}$$

($\because$ for Reflected current has 180° out of phase)

Where V_r, V_i are Reflected and Incident Voltages.

I_r, I_i are Reflected and Incident Currents.

so $\boxed{\Gamma = \frac{V_r}{V_i} = \frac{Z_R - Z_0}{Z_R + Z_0}}$

Input Impedance in terms of Reflection Coefficient:-

We know that the input impedance of a transmission line terminated with Z_R is

$$Z_{in} = Z_0 \left(\frac{Z_R + Z_0 \tanh \Gamma l}{Z_0 + Z_R \tanh \Gamma l} \right)$$

$$\Rightarrow Z_{in} = Z_0 \left[\frac{Z_R \cosh \Gamma l + Z_0 \sinh \Gamma l}{Z_0 \cosh \Gamma l + Z_R \sinh \Gamma l} \right]$$

In exponential form,

$$\cosh \Gamma l = \frac{e^{\Gamma l} + e^{-\Gamma l}}{2}, \quad \sinh \Gamma l = \frac{e^{\Gamma l} - e^{-\Gamma l}}{2}$$

Thus.

$$Z_{in} = Z_0 \left[\frac{Z_R (e^{\Gamma l} + e^{-\Gamma l}) + Z_0 (e^{\Gamma l} - e^{-\Gamma l})}{Z_0 (e^{\Gamma l} + e^{-\Gamma l}) + Z_R (e^{\Gamma l} - e^{-\Gamma l})} \right] \cdot \frac{(\cancel{2})}{(\cancel{2})}$$

$$= Z_0 \left[\frac{e^{\Gamma l} (Z_R + Z_0) + e^{-\Gamma l} (Z_R - Z_0)}{e^{\Gamma l} (Z_0 + Z_R) - e^{-\Gamma l} (Z_R - Z_0)} \right]$$

$$= Z_0 \left[\frac{\cancel{e^{\Gamma l} (Z_R + Z_0)} \left\{ 1 + \frac{e^{-\Gamma l} (Z_R - Z_0)}{e^{\Gamma l} (Z_R + Z_0)} \right\}}{\cancel{e^{\Gamma l} (Z_R + Z_0)} \left\{ 1 - \frac{e^{-\Gamma l} (Z_R - Z_0)}{e^{\Gamma l} (Z_R + Z_0)} \right\}} \right]$$

$$\therefore Z_{in} = Z_0 \left[\frac{1 + e^{-2\Gamma l} \cdot \Gamma}{1 - e^{-2\Gamma l} \cdot \Gamma} \right]$$

Where $\Gamma = \frac{Z_R - Z_0}{Z_R + Z_0}$

3.1 Coulomb's law

Coulomb's law states that the force between two point charges Q_1 and Q_2 is directly proportional to the product of the charges and inversely proportional to the square of the distance R , between them.

Expressed mathematically,

$$F \propto \frac{Q_1 Q_2}{R^2}$$



$$F = \frac{k Q_1 Q_2}{R^2} \quad \text{--- (1)}$$

where k is the proportionality constant.

In SI units,

charges Q_1 and Q_2 are expressed in Coulombs (C)

distance R is in meters (m)

force F in Newtons (N), and

$$k = \frac{1}{4\pi\epsilon_0} \approx 9 \times 10^9 \text{ N/m}^2 \text{C}^2$$

where ϵ_0 = permittivity of free space (farads/metre)

$$= 8.85 \times 10^{-12} \approx \frac{10^{-9}}{36\pi} \text{ F/m}$$

$\therefore$ eqn(1) becomes

$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \text{ (N)} \quad \text{--- (2)}$$

Coulomb's Law in Vector Form:

As shown in fig 1., the point charges Q_1 and Q_2 are located at points having position vectors $\vec{r}_1$ and $\vec{r}_2$. Then the force $\vec{F}_{12}$ on Q_2 due to Q_1 is given by

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \hat{a}_{R_{12}}$$

Where,

the distance vector between two charges is

$$\vec{R}_{12} = \vec{r}_2 - \vec{r}_1 \quad \text{---(2)}$$

$$\Rightarrow R = |\vec{R}_{12}| = |\vec{r}_2 - \vec{r}_1| \quad \text{---(3)}$$

and Unit vector $\hat{a}_{R_{12}} = \frac{\vec{R}_{12}}{|\vec{R}_{12}|} = \frac{\vec{R}_{12}}{R} \quad \text{---(4)}$

By substituting eq(4) in eq(1), we rewrite eq(1) as

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R} \cdot \frac{\vec{R}_{12}}{R}$$

$$\vec{F}_{12} = \frac{Q_1 Q_2 \cdot \vec{R}_{12}}{4\pi\epsilon_0 R^3} \quad \text{---(5)}$$

(or) from eq(3), eq(5) gives

$$\vec{F}_{12} = \frac{Q_1 Q_2 (\vec{r}_2 - \vec{r}_1)}{4\pi\epsilon_0 |\vec{r}_2 - \vec{r}_1|^3} \quad \text{---(6)}$$

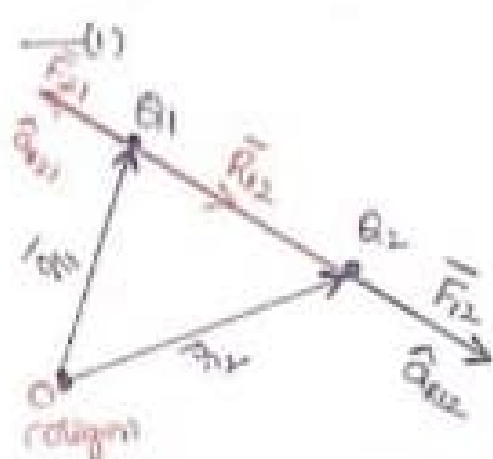


Fig) Coulomb vector between point charges Q_1 and Q_2 .

Question 1: Point charges 1mC and -2mC are located at $\mathbf{A}(3, 2, -1)$ and $\mathbf{B}(-1, -1, 4)$, respectively. Calculate the electric force on a 10nC charge located at $\mathbf{C}(0, 3, 1)$ and the electric field intensity at that point.

Solution: Given that

$$Q_1 = 1\text{mC} = 1 \times 10^{-3}\text{C} \text{ at } (3, 2, -1), \quad \vec{r}_1 = 3\hat{a}_x + 2\hat{a}_y - \hat{a}_z$$

$$Q_2 = -2\text{mC} = -2 \times 10^{-3}\text{C} \text{ at } (-1, -1, 4), \quad \vec{r}_2 = -\hat{a}_x - \hat{a}_y + 4\hat{a}_z$$

$$Q = 10\text{nC} = 10 \times 10^{-9}\text{C} \text{ at } (0, 3, 1), \quad \vec{r} = 3\hat{a}_y + \hat{a}_z$$

$\therefore$ Using Coulomb's law, and the superposition principle, the force on Q at position vector $\vec{r}$ is

$$\vec{F} = \frac{Q}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k (\vec{r} - \vec{r}_k)}{|\vec{r} - \vec{r}_k|^3}$$

$$k=1, 2$$

$$\therefore \vec{F} = \frac{Q}{4\pi\epsilon_0} \left[\frac{Q_1 (\vec{r} - \vec{r}_1)}{|\vec{r} - \vec{r}_1|^3} + \frac{Q_2 (\vec{r} - \vec{r}_2)}{|\vec{r} - \vec{r}_2|^3} \right] \quad \text{--- (1)}$$

$$\therefore \text{the distance vector } \vec{r} - \vec{r}_1 = (0, 3, 1) - (3, 2, -1) = (-3, 1, 2)$$

$$(\text{or}) \vec{r} - \vec{r}_1 = -3\hat{a}_x + \hat{a}_y + 2\hat{a}_z$$

$$\therefore \vec{r} - \vec{r}_2 = (0, 3, 1) - (-1, -1, 4) = (1, 4, -3)$$

$$\therefore \vec{F} = \frac{10 \times 10^{-9}}{4\pi \times 10^{-9} \times 36\pi} \left[\frac{1 \times 10^{-3} (-3, 1, 2)}{|(-3, 1, 2)|^3} - \frac{2 \times 10^{-3} (1, 4, -3)}{|(1, 4, -3)|^3} \right]$$

$$\vec{F} = \frac{10 \times 10 \times 10^{-9}}{4\pi \times 10^{-9} \times \frac{36\pi}{9}} \left[\frac{(-3, 1, 2)}{[9+1+4]^{3/2}} - \frac{2(1, 4, -3)}{[1+16+9]^{3/2}} \right] \quad (16)$$

$$\vec{F} = 10^{-3} \times 9 \times 10 \left[\frac{(-3, 1, 2)}{14\sqrt{14}} - \frac{(2, 8, -6)}{26\sqrt{26}} \right]$$

$$\vec{F} = 90 \times 10^{-3} \left[0.019(-3, 1, 2) - 7.54 \times 10^{-3}(2, 8, -6) \right]$$

$$\vec{F} = 10^{-3} \left[1.71(-3, 1, 2) - 0.6786(2, 8, -6) \right]$$

$$= 10^{-3} \left[(-5.13, 1.71, 3.42) - (1.357, 5.4288, -4.0716) \right]$$

$$= 10^{-3} \left[(-6.51, -3.718, 7.50) \right]$$

$$\boxed{\vec{F} = -6.51 \hat{a}_x - 3.718 \hat{a}_y + 7.50 \hat{a}_z} \text{ mN}$$

At point. $(0, 3, 1)$

$$\therefore \text{Electric field } \vec{E} = \frac{\vec{F}}{q} = \frac{(-6.51 \hat{a}_x - 3.718 \hat{a}_y + 7.50 \hat{a}_z) \times 10^{-3}}{10 \times 10^{-9}}$$

$$\vec{E} = 65$$

$$\boxed{\vec{E} = -651.0 \hat{a}_x - 371.8 \hat{a}_y + 750 \hat{a}_z} \text{ KV/m}$$

$$\therefore \oint_S \vec{D} \cdot d\vec{S} = Q_{enc} = \psi \quad \text{Hence proved.}$$

Gauss law in point form:- [Maxwell's 1st Equation]

Statement:- "It states that the divergence of electric flux density is equal to volume charge density at a given point in a medium."

$$\boxed{\nabla \cdot \vec{D} = \rho_v} \quad \text{or} \quad \boxed{\text{div } \vec{D} = \rho_v}$$

Proof:-

We know that Gauss law in integral form is

$$\oint_S \vec{D} \cdot d\vec{S} = Q_{enc} = \psi \rightarrow (1)$$

But volume charge density is

$$\rho_v = \frac{dq}{dv} \Rightarrow dq = \rho_v dv$$

$$Q = \int_V \rho_v dv \rightarrow (2)$$

from (1) and (2)

$$\oint_S \vec{D} \cdot d\vec{S} = \int_V \rho_v dv \rightarrow (3)$$

Applying divergence theorem on eq (3)

$$\therefore \oint_S \vec{D} \cdot d\vec{S} = \int_V (\nabla \cdot \vec{D}) dv \rightarrow (4)$$

Comparing eq (3) and (4) we get

$$\therefore \boxed{\nabla \cdot \vec{D} = \rho_v} \quad \text{Hence proved.}$$

Applications of Gauss law:-

By using Gauss law we have to calculate

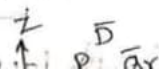
- (i) Electric flux density ($\vec{D}$)
- (ii) Electric field intensity ($\vec{E}$)
- (iii) charge enclosed by the surface (Q)

The Gauss law can be applied to point charge, line charge, surface charge, volume charge.

Limitation:- Gauss law can not be applicable for non-gaussian surface (or) un-symmetrical surfaces.

Gauss law due to a point charge

Let us consider a point charge q located at origin



PRACTICE EXERCISE 7.4

If the solenoid of Figure 7.9 has 2,000 turns, a length of 75 cm, a radius of 5 cm, and carries a current of 50 mA along $\mathbf{a}_\phi$, find $\mathbf{H}$ at

- (a) (0, 0, 0)
- (b) (0, 0, 75 cm)
- (c) (0, 0, 50 cm)

Answer: (a) $66.52\mathbf{a}_z$ A/m, (b) $66.52\mathbf{a}_z$ A/m, (c) $131.7\mathbf{a}_z$ A/m.

E'S CIRCUIT LAW—MAXWELL'S EQUATION

Ampere's circuit law states that the line integral of the tangential component of $\mathbf{H}$ around a *closed* path is the same as the net current I_{enc} enclosed by the path.

In other words, the circulation of $\mathbf{H}$ equals I_{enc} ; that is,

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}} \quad (7.16)$$

Ampere's law is similar to Gauss's law and it is easily applied to determine $\mathbf{H}$ when the current distribution is symmetrical. It should be noted that eq. (7.16) always holds whether the current distribution is symmetrical or not but we can only use the equation to determine $\mathbf{H}$ when symmetrical current distribution exists. Ampere's law is a special case of Biot-Savart's law; the former may be derived from the latter.

By applying Stoke's theorem to the left-hand side of eq. (7.16), we obtain

$$I_{\text{enc}} = \oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S} \quad (7.17)$$

Downloaded From : www.EasyEngineering.net

Downloaded From : www.EasyEngineering.net

Magnetostatic Fields

But

$$I_{\text{enc}} = \int_S \mathbf{J} \cdot d\mathbf{S} \quad (7.18)$$

Comparing the surface integrals in eqs. (7.17) and (7.18) clearly reveals that

$$\nabla \times \mathbf{H} = \mathbf{J} \quad (7.19)$$

This is the third Maxwell's equation to be derived; it is essentially Ampere's law in differential (or point) form whereas eq. (7.16) is the integral form. From eq. (7.19), we should observe that $\nabla \times \mathbf{H} = \mathbf{J} \neq 0$; that is, magnetostatic field is not conservative.

APPLICATIONS OF AMPERE'S LAW

We now apply Ampere's circuit law to determine $\mathbf{H}$ for some symmetrical current distributions as we did for Gauss's law. We will consider an infinite line current, an infinite

8.

Types of charge distribution

Line charge

Surface charge

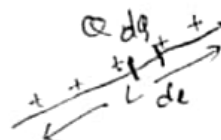
volume charge

Line charge distribution: charge distributed per unit length is known as line charge density. (ρ_L), C/m.

charge per unit length is line charge.

$$\rho_L = \frac{dQ}{dl} \text{ C/m.}$$

$$dQ = \rho_L \cdot dl.$$



Total charge, $Q = \int dQ$.

$$\int dQ = \int \rho_L dl.$$

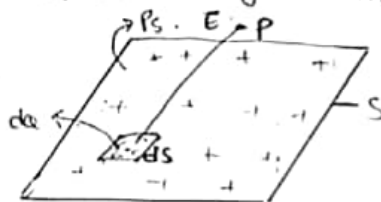
$$Q = \int_L \rho_L \cdot dl.$$

surface charge distribution:

charge distributed per unit area is known as surface charge density.

$$\rho_S = \frac{dQ}{dS} \text{ C/m}^2$$

$$dQ = \rho_S \cdot dS.$$



Total charge, $Q = \int dQ$.

$$\int_S dQ = \int_S \rho_S \cdot dS.$$

$$\therefore \int_S = \iint.$$

$$Q = \int_S \rho_S \cdot dS.$$

$$\rho_S = \frac{\text{charge}}{\text{unit area}} = \frac{dQ}{dS} = \text{C/m}^2$$

volume charge distribution:

charge distributed per unit volume is known as volume charge density.

$$\rho_V = \frac{\text{charge}}{\text{unit volume}} = \frac{dQ}{dV} \text{ C/m}^3.$$

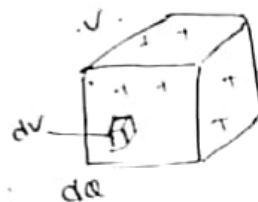
$$\rho_V = \frac{dQ}{dV} \text{ C/m}^3.$$

$$dQ = \rho_V \cdot dV.$$

Total charge, $Q = \int dQ$

$$\int_V dQ = \int_V \rho_V \cdot dV$$

$$Q = \int_V \rho_V \cdot dV$$



$$\therefore \int_V = \iiint$$

UNIT IV

Maxwell's Equations (Time Varying Fields):

- Faraday's Law and Transformer EMF,
- Inconsistency of Ampere's Law and
- Displacement Current Density,
- Maxwell's Equations in Different Final Forms and
- Word Statements .
- Conditions at a Boundary Surface.
- Illustrative Problems.

4.1 Faraday's Law of Induction:

According to Faraday's experiment a voltage known as **electromotive force**(or **emf**) is induced in a closed circuit due to time-varying magnetic fields which causes a flow of current. It is denoted by V_{emf} and measured in Volts. This Induced emf, i.e V_{emf} is equal to the time rate of change of the magnetic flux linkage in any closed circuit. This is known as **Faraday's Law**.

Mathematically it is expressed as:

$$V_{\text{emf}} = - \frac{d\lambda}{dt} = - N \frac{d\psi}{dt} \dots\dots\dots(1)$$

Where λ is flux linkage, Ψ is flux passing through each turn N is the number of turns in a circuit. The negative sign indicates that induced emf is in a direction such that the current generated by it opposes the magnetic flux producing it. This statement is known as **Lenz's Law**.

Consider the electric circuit of fig 4.1, the battery is a emf of source. The electrochemical action of the battery results in an emf-produced field $\mathbf{E}_f$. Due to the accumulation of charge at battery terminals, an electrostatic field $\mathbf{E}_e (= -\nabla V)$ also exists.

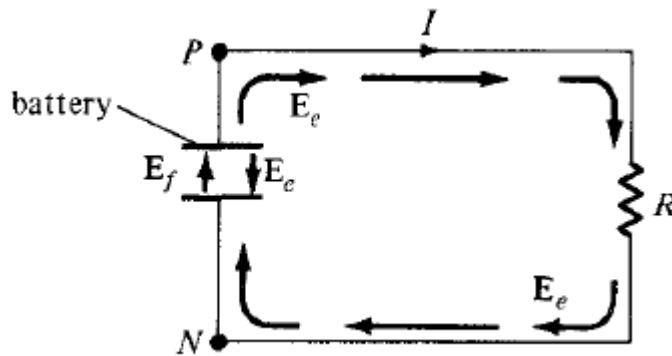


Fig4.1 A circuit showing emf-produced field E_f and Electrostatic field E_e

The total electric field at any point is

$$\mathbf{E} = \mathbf{E}_f + \mathbf{E}_e \quad \dots\dots\dots(2) \quad \because \quad \mathbf{E}_f = - \mathbf{E}_e$$

Note:

- $\mathbf{E}_f$ is zero outside the battery
- $\mathbf{E}_f$ and $\mathbf{E}_e$ have opposite directions in the battery, and the direction of $\mathbf{E}_e$ inside the battery is opposite to that outside it

Now integrate equation(2) over the closed circuit

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = \oint_L \mathbf{E}_f \cdot d\mathbf{l} + 0 = \int_N^P \mathbf{E}_f \cdot d\mathbf{l} \quad (\text{through battery}) \quad (3)$$

where $\oint \mathbf{E}_e \cdot d\mathbf{l} = 0$ because $\mathbf{E}_e$ is conservative. The emf of the battery is the line integral of the emf-produced field; that is,

$$V_{\text{emf}} = \int_N^P \mathbf{E}_f \cdot d\mathbf{l} = - \int_N^P \mathbf{E}_e \cdot d\mathbf{l} = IR \quad (4)$$

The induced emf expression in equation (1) can be expressed in terms of Electric field intensity $\mathbf{E}$,

$$V_{\text{emf}} = \oint_l \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} \quad \dots\dots\dots(5)$$

Where l is the length of the closed path. In electrostatics, this relation leads to zero potential difference as electric field is conservative. However in time varying fields, this relation leads to the induced emf or voltage about specific closed path.

4.2 Transformer and Motional emfs

Faraday's law established the link between the magnetic and electric fields. For $N=1$, the eq.(1) can be written as

$$V_{\text{emf}} = - \frac{d\psi}{dt} \dots\dots\dots(3)$$

We know that flux Ψ is expressed as surface integral of magnetic flux density $\mathbf{B}$ as:

$$\psi = \oint_s \vec{\mathbf{B}} \cdot \vec{ds} \dots\dots\dots(4)$$

Combining the equations (2), (3) and (4), we get

$$V_{\text{emf}} = \oint_l \vec{E} \cdot \vec{dl} = - \frac{d}{dt} \int_s \vec{B} \cdot \vec{ds} \dots\dots\dots(5)$$

Consider three different cases for the generation of time-varying flux:

- Stationary loop in a time-varying magnetic field $\mathbf{B}$ (**Transformer Emf**)
- Moving loop area in a static magnetic field $\mathbf{B}$ (**Motional Emf**)
- Moving loop area in a time-varying magnetic field $\mathbf{B}$

4.2.1 Stationary loop in a time-varying magnetic field **B** (Transformer emf)

Consider a conducting path in a time-varying magnetic field **B** which is increasing with time *t* as shown in fig. 4.2. Then the eq.(5) becomes

$$V_{\text{emf}} = \oint_l \vec{E} \cdot d\vec{l} = - \int_s \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s} \quad \dots(6)$$

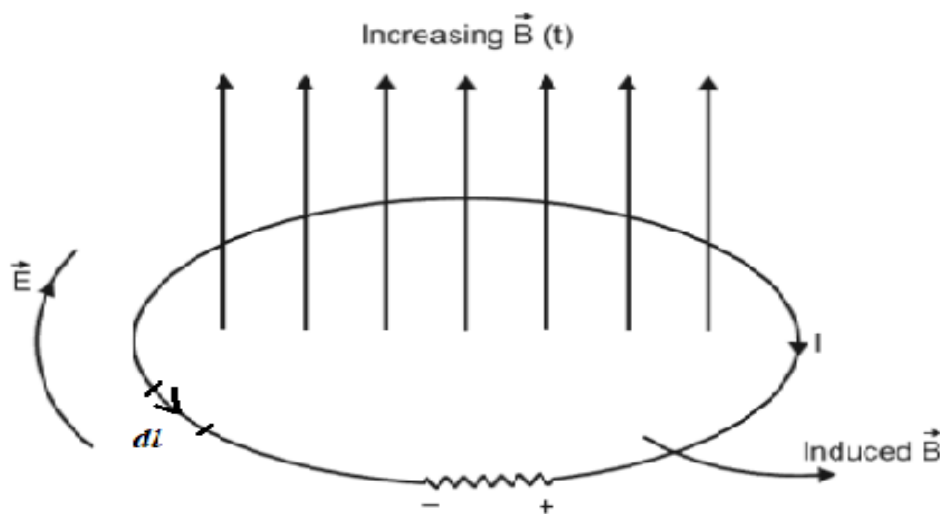


Fig 4.2 Induced emf in stationary loop in a time-varying magnetic field **B**

This induced emf by the time-varying current (producing the time-varying **B** – field) in a stationary loop is referred to as **Transformer emf** in power analysis since it is due to transformer action.

Using Stokes's theorem to the middle term of eq.(6), we obtain

$$\begin{aligned} \oint_l \vec{E} \cdot d\vec{l} &= \int_s \nabla \times \vec{E} \cdot d\vec{s} \\ \int_s \nabla \times \vec{E} \cdot d\vec{s} &= - \int_s \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s} \quad \dots(7) \end{aligned}$$

Thus we get, $\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad \dots\dots(8)$

Equation (8) is one of the Maxwell's equations in time-varying fields. Here, $\nabla \times \vec{E} \neq 0$, this implies that the time-varying field is not conservative.

4.2.2 Moving loop in static field(Motional emf or Generator emf)

When a conducting loop is moving in a static $\vec{B}$ field , an emf is induced in a loop. Recalling, the equation for the force on a charge moving uniform velocity $\vec{v}$ in a magnetic field $\vec{B}$, the magnetic force $\vec{F}_M$ is given as:

$$\vec{F}_M = Q\vec{v} \times \vec{B} \dots\dots(9)$$

Then the motional electric field $\vec{E}_M$ as

$$\vec{E}_M = \frac{\vec{F}_M}{Q} = \vec{v} \times \vec{B} \dots\dots(10)$$

Now, the emf induced in closed path consisting of large number of free electrons, moving with the velocity $\vec{v}$ is given from eq.(5) and by substituting eq.(10) as:

$$V_{\text{emf}} = \oint_l \vec{E}_M \cdot d\vec{l} = \oint_l \vec{v} \times \vec{B} \dots\dots(11)$$

This emf is often referred to as ***motional emf or generator emf or flux- cutting emf.***

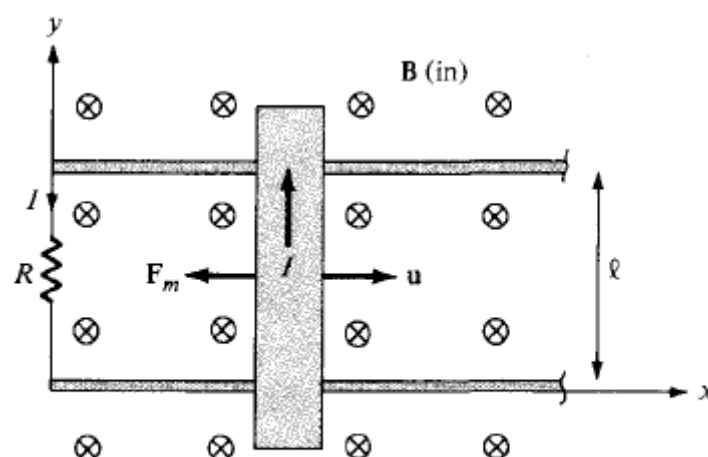


Figure 4.2 Induced emf due to a moving loop in a static $\vec{B}$ field.

Applying Stoke's theorem to eq.(11), we get

$$\int_S (\nabla \times \vec{E}_M) \cdot d\vec{s} = \int_S \nabla \times (\vec{v} \times \vec{B}) \cdot d\vec{s}$$

or

$$\nabla \times \vec{E}_M = \nabla \times (\vec{v} \times \vec{B}) \dots (12)$$

Negative sign is not included as per Lenz's law

4.2.3 Moving loop area in a time-varying magnetic field B

In this case, the effects of both time-varying field and time-varying path are considered. Here, both transformer and motional emf are present. Hence, the total emf is obtained on summing Equations (6) and (11), we get:

$$V_{emf} = \oint_l \vec{E} \cdot d\vec{l} = - \int_s \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s} + \oint_l (\vec{u} \times \vec{B}) \cdot d\vec{l} \dots (13)$$

Applying Stokes's theorem in the above equation, we get:

$$\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} + \nabla \times (\vec{u} \times \vec{B}) \dots (14)$$

Note: The field generated by the induced emf V_{emf} is non-conservative. It depends on the path chosen.

4.3 Inconsistency in Ampere's Law for Time varying fields

Maxwell's curl equation for electrostatic fields and modified it for time-varying situations to satisfy Faraday's law. Reconsider Maxwell's curl equation for magnetic fields (Ampere's circuit law) for time-varying conditions.

For static EM fields, we recall that

$$\nabla \times \vec{H} = \vec{J} \dots\dots\dots(15)$$

But the divergence of the curl of any vector field is identically zero ,(that is $\nabla \cdot \nabla \times \vec{A} = 0$)

Hence,

$$\nabla \cdot (\nabla \times \vec{H}) = 0 = \nabla \cdot \vec{J} \dots\dots\dots(16)$$

Now, from the continuity equation we have:

$$\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t} \neq 0 \dots\dots\dots(17)$$

Hence, for time-varying fields, Equation (15) must be modified so as to satisfy Equation (17) as:

$$\nabla \times \vec{H} = \vec{J} + \vec{J}_d \dots\dots\dots(18)$$

Here, in this equation we have added a new term $\vec{J}_d$ which is not known and is to be determined. To determine $\vec{J}_d$ we take divergence on both sides of Equation (18), we get:

$$\nabla \cdot (\nabla \times \vec{H}) = 0 = \nabla \cdot \vec{J} + \nabla \cdot \vec{J}_d \dots\dots\dots(19)$$

Combining Equations (17) and (19), we get:

$$\nabla \cdot \vec{J}_d = -\nabla \cdot \vec{J} = \frac{\partial \rho_v}{\partial t} \dots\dots\dots(20)$$

We know that the volume charge density ρ_v is related to electric flux density $\vec{D}$ as:

$$\nabla \cdot \vec{D} = \rho_v$$

Substituting this value in Equation (20), we get:

$$\nabla \cdot \vec{J}_d = \frac{\partial}{\partial t} (\nabla \cdot \vec{D}) = \nabla \cdot \frac{\partial \vec{D}}{\partial t}$$

Thus, the unknown term $\vec{J}_d$ is obtained as:

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$$

Hence, Equation (18) can be written as:

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \dots\dots\dots(21)$$

Hence eq.(21) is the Modified Ampere's law for Time varying fields. The term $\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$ is known as displacement current density and $\vec{J}$ is the conduction current density ($\vec{J} = \sigma \vec{E}$).

Displacement current

Equation (21) represents Maxwell's curl equation for time-varying fields in differential form where $\vec{J}_d$ is called as **displacement current density** and $\vec{J}$ as **conduction current density** given as $\vec{J} = \sigma \vec{E}$ which is due to the motion of charges.

Now, we can define **displacement current** I_d through a given surface s as surface integral of displacement current density $\vec{J}_d$ as:

$$I_d = \int \vec{J}_d \cdot d\vec{s} = \int \frac{\partial \vec{D}}{\partial t} \cdot d\vec{s}$$

The displacement current is generated due to time-varying electric field only.

Applying Stokes's theorem, Equation (21) we get:

$$\oint_l \vec{H} \cdot d\vec{l} = \int_s \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s} = I + I_d \quad \dots(22)$$

[Since $\int_s \vec{J} \cdot d\vec{s} = I$]

4.4 Maxwell's Equations in Different Final Forms and Word Statements

Table 4.1 *Maxwell's Equations for Time varying Em fields*

Sl. No.	Differential Form	Integral Form	Name
1	$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$	$\oint_l \vec{H} \cdot d\vec{l} = \int_s \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$	Modified Ampere's circuital law
2	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint_l \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \int_s \vec{B} \cdot d\vec{s}$	Faraday's law of electromagnetic induction
3	$\nabla \cdot \vec{D} = \rho_v$	$\oint_s \vec{D} \cdot d\vec{s} = \oint_v \rho_v dv = Q$	Gauss' law of Electrostatics
4	$\nabla \cdot \vec{B} = 0$	$\oint_s \vec{B} \cdot d\vec{s} = 0$	Gauss' law of Magnetostatic (non-existence of magnetic mono-pole)

Word Statements

$$1. \quad \nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad \text{or,} \quad \oint_l \vec{H} \cdot d\vec{l} = \int_s \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{s}$$

The magnetomotive force around a closed path is equal to the conduction current plus the time derivative of the electric displacement through any surface bounded by the path. This is the modified Ampere's law for time varying fields.

$$2. \quad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{or,} \quad \oint_l \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \int_s \vec{B} \cdot d\vec{s}$$

The electromotive force around a closed path is equal to the time derivative of the magnetic displacement through any surface bounded by the path. This is the Faraday's law for electromagnetic fields.

$$3. \quad \nabla \cdot \vec{D} = \rho_v \quad \text{or,} \quad \oint_s \vec{D} \cdot d\vec{s} = \oint_v \rho_v dv = Q$$

The total electric displacement through any closed surface enclosing a volume is equal to the total charge within the volume. This is the Gauss' law for static electric fields.

$$4. \quad \nabla \cdot \vec{B} = 0 \quad \text{or,} \quad \oint_s \vec{B} \cdot d\vec{s} = 0$$

The net magnetic flux emerging through any closed surface is zero. In other words, the magnetic flux lines do not originate and end anywhere, but are continuous. This is the Gauss' law for static magnetic fields, which confirms the non-existence of magnetic monopole.

4.5 Maxwell's Equations for Time-Harmonic fields(Phasor Form)

So far, our time dependence of EM fields has been arbitrary. To be specific, we shall assume that the fields are *time harmonic*.

A **time-harmonic field** is one that varies periodically or sinusoidally with time.

In general, a phasor can be a scalar or a vector. If a vector $\vec{A}(x, y, z, t)$ is a time-harmonic field, the phasor form of $\vec{A}$ is $\bar{A}(x, y, z)$; the relation being

$$\boxed{\vec{A} = \text{Re}[\bar{A}e^{j\omega t}]}$$

$$\therefore \quad \vec{A}(x, y, z, t) = \text{Re}[\bar{A}(x, y, z)e^{j\omega t}]$$

$$\boxed{\frac{\partial \bar{A}}{\partial t} \rightarrow j\omega \bar{A}} \quad \text{and} \quad \boxed{\int \bar{A} dt \rightarrow \frac{\bar{A}}{j\omega}}$$

To transform the instantaneous Maxwell's equations into time-harmonic forms, we replace all sources and field quantities by their phasor equivalents and replace all time-derivatives of quantities with $j\omega$ times the phasor equivalent. Thus, the Maxwell's equations for time-harmonic fields are given in Table 4.2.

Sl. No.	Differential Form	Integral Form	Name
1	$\nabla \times \bar{H} = \bar{J} + j\omega\bar{D}$	$\oint_l \bar{H} \cdot d\bar{l} = \int_s (\bar{J} + j\omega\bar{D}) \cdot d\bar{s}$	Modified Ampere's circuital law
2	$\nabla \times \bar{E} = -j\omega\bar{B}$	$\oint_l \bar{E} \cdot d\bar{l} = -j\omega \int_s \bar{B} \cdot d\bar{s}$	Faraday's law of electromagnetic induction
3	$\nabla \cdot \bar{D} = \rho_v$	$\oint_s \bar{D} \cdot d\bar{s} = \int_v \rho_v dv$	Gauss' law of Electrostatic
4	$\nabla \cdot \bar{B} = 0$	$\oint_s \bar{B} \cdot d\bar{s} = 0$	Gauss' law of Magnetostatic (non-existence of magnetic mono-pole)

4.6 Conditions at a Boundary Surface

If the field exists in a region consisting of two different media, the conditions that the field must satisfy at the interface separating the media are called **boundary conditions**. These conditions are helpful in determining the field on one side of the boundary if the field on the other side is known. The electromagnetic fields that are solved using Maxwell's equations must satisfy boundary conditions at the interface between different media.

The boundary conditions on electric and magnetic fields at any surface of discontinuity are given by:

1. The tangential component of electric field, E is continuous across any discontinuity, that is,

$$E_{\tan 1} = E_{\tan 2}$$

Subscript tan1 is the tangential component of the field at the boundary in medium 1. Subscript tan2 represents it in medium 2.

2. The tangential component of magnetic field, H is continuous across any surface except at the surface of a perfect conductor.

$$H_{\tan 1} = H_{\tan 2}$$

At the surface of a perfect conductor, the tangential component of H is discontinuous by quantity equal to the surface current density K (A/m).

$$H_{\tan 1} - H_{\tan 2} = K$$

3. The normal component of D is continuous except at a surface which has surface charge density.

$$D_{n1} = D_{n2}$$

At the surface where surface charge density exists, the normal component of D is discontinuous by a quantity equal to the surface charge density.i.e

$$D_{n1} - D_{n2} = \rho_s$$

4.The normal component of B is continuous across any discontinuity.

$$B_{n1} = B_{n2}$$

4.6.1 Boundary conditions for Electric fields

- ❖ We shall consider the boundary conditions at an interface separating
 - Dielectric (ϵ_{r1}) and Dielectric (ϵ_{r2})
 - Conductor and dielectric
 - Conductor and free space
- ❖ To determine the boundary conditions, we need to use Maxwell's equations:

$$\oint_l \vec{E} \cdot d\vec{l} = 0 \quad \dots(1)$$

and

$$\oint_s \vec{D} \cdot d\vec{s} = Q_{enc} \quad \dots(2)$$

- ❖ The electric field intensity $\vec{E}$ is decomposed into two orthogonal components:

$$\vec{E} = \vec{E}_t + \vec{E}_n \quad \dots\dots\dots(3)$$

where $\vec{E}_t$, and $\vec{E}_n$ are, respectively, the tangential and normal components of $\vec{E}$ to the interface of interest. A similar decomposition can be done for the electric flux density $\vec{D}$.

A. Dielectric-Dielectric Boundary Conditions

Consider the $\mathbf{E}$ -field existing in a region consisting of two different dielectrics characterized by $\epsilon_1 = \epsilon_o \epsilon_{r1}$ and $\epsilon_2 = \epsilon_o \epsilon_{r2}$ as shown in Figure 4.3(a). $\mathbf{E}_1$ and $\mathbf{E}_2$ in media 1 and 2, respectively, can be decomposed as

$$\mathbf{E}_1 = \mathbf{E}_{1t} + \mathbf{E}_{1n} \quad \text{.....4(a)}$$

$$\mathbf{E}_2 = \mathbf{E}_{2t} + \mathbf{E}_{2n} \quad \text{.....4(b)}$$

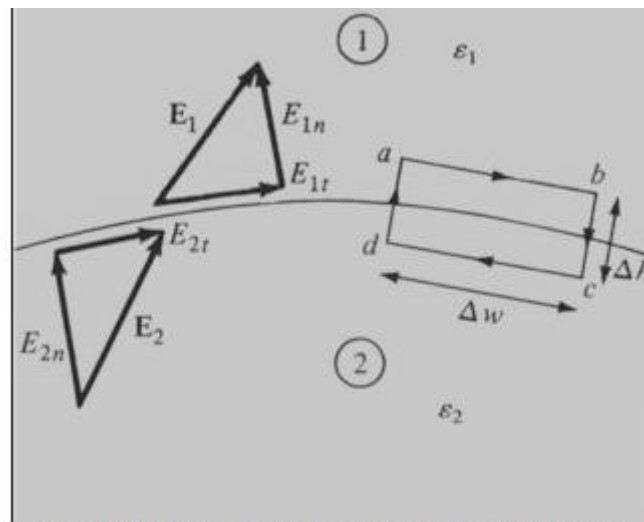


Fig. 4.3(a) Tangential components of $\mathbf{E}$ -field

- ❖ To obtain the boundary conditions on the **tangential components** of $\vec{\mathbf{E}}$ across two boundaries, apply Maxwell's equation (1), around a closed path $a-b-c-d-a$, as shown in fig 4.3(a).

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = \int_a^b \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} + \int_b^c \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} + \int_c^d \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} + \int_d^a \vec{\mathbf{E}} \cdot d\vec{\mathbf{l}} = 0$$

$$E_{1t} \Delta w - E_{1n} \frac{\Delta h}{2} - E_{2n} \frac{\Delta h}{2} - E_{2t} \Delta w + E_{2n} \frac{\Delta h}{2} + E_{1n} \frac{\Delta h}{2} = 0 \quad \text{...(5)}$$

as $\Delta h \rightarrow 0$, eq. (5) becomes

$$E_{1t} \Delta w - E_{2t} \Delta w = 0$$

$$\boxed{E_{1t} = E_{2t}} \quad \text{.....(6)}$$

- ❖ Thus the tangential components of $\mathbf{E}$ are the same on the two sides of the boundary. In other words, $\mathbf{E}_t$ undergoes no change on the boundary and it is said to be **continuous across the boundary**.

❖ Since $\mathbf{D} = \epsilon \mathbf{E} = \mathbf{D}_t + \mathbf{D}_n$, then eq. (6) becomes

$$E_{1t} = \frac{D_{1t}}{\epsilon_1} \text{ and } E_{2t} = \frac{D_{2t}}{\epsilon_2}$$

Thus, Equation (45) can be rewritten as:

$$\frac{D_{1t}}{\epsilon_1} = \frac{D_{2t}}{\epsilon_2} \quad \dots (7)$$

Equation (7) shows that the tangential component of electric flux density $\vec{D}$ is not same across the interface, that is, D_t is discontinuous across the boundary.

❖ Now, Let us derive boundary conditions for **Normal components** of $\mathbf{E}$ and $\mathbf{D}$.

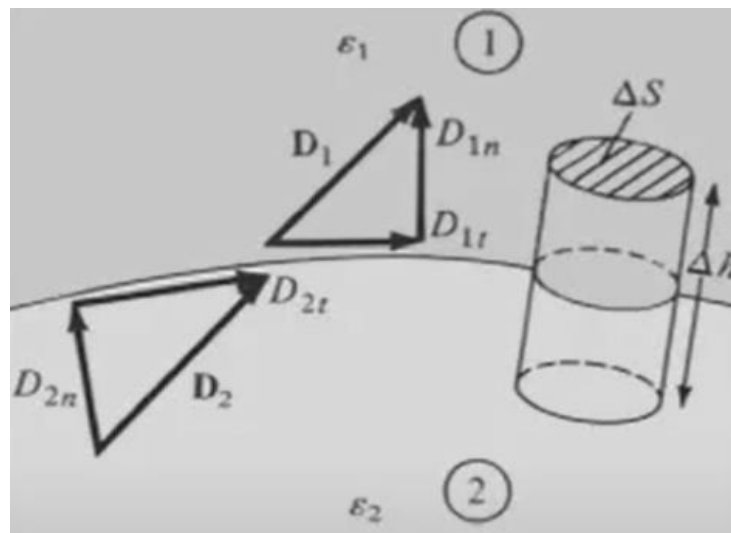


Fig. 4.3(b) Normal components of E-field

❖ To derive boundary conditions for normal components, apply Gauss's law in integral form (eq.2) to the pill box spreading over the interface from medium 1 to medium 2 as shown in fig. 4.3(b)

$$\oint_s \vec{D} \cdot d\vec{s} = Q_{\text{enc}}$$

$$\int_{\text{top}} \vec{D} \cdot d\vec{s} + \int_{\text{bottom}} \vec{D} \cdot d\vec{s} + \int_{\text{sides}} \vec{D} \cdot d\vec{s} = \Delta Q = \rho_s \Delta S$$

If we allow $\Delta h \rightarrow 0$, the above equation becomes

$$D_{1n} \Delta s - D_{2n} \Delta s = \rho_s \Delta s$$

$$D_{1n} - D_{2n} = \rho_s$$

If there are no free charges present at the boundary, then $\rho_s = 0$. This gives:

$$D_{1n} = D_{2n} \quad \dots(8)$$

$$\epsilon_1 E_{1n} = \epsilon_2 E_{2n} \quad \dots(9)$$

Equation (8) clearly shows that the normal component of the flux density $\vec{D}$ is the same on either side of the boundary, whereas equation (9) shows that the normal component of the electric field intensity $\vec{E}$ varies across the boundary.

Equations (6) to (9) are collectively known as **boundary conditions** which must be satisfied by an electric field that exists at a boundary separating two dielectric media.

B. Conductor – dielectric boundary conditions

This is the case shown in Figure 4.4. The conductor is assumed to be perfect (i.e., $\sigma \rightarrow \infty$ or $\rho_c \rightarrow 0$)

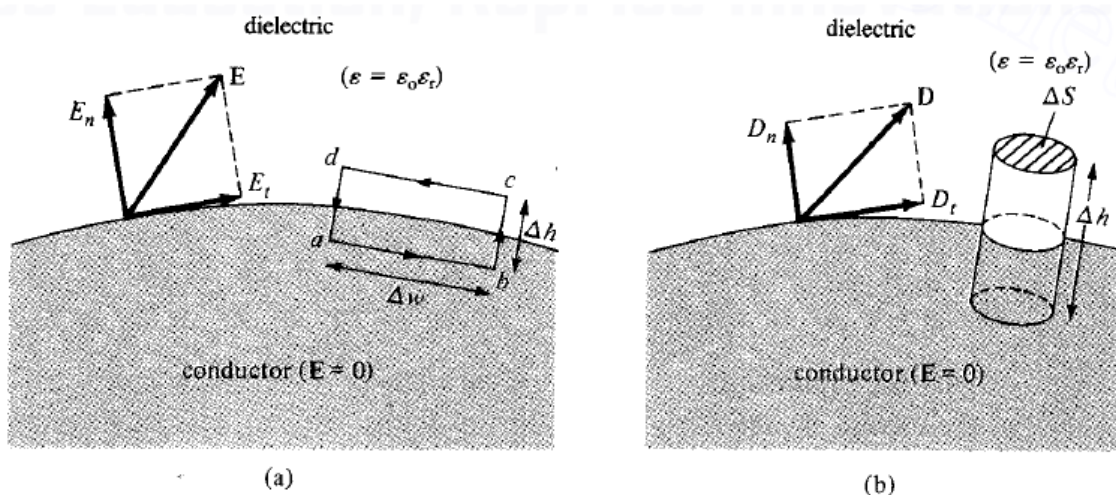


Figure 4.4 Conductor–dielectric boundary.

To obtain the boundary conditions on the **tangential components** of $\vec{E}$ across two boundaries, apply Maxwell's equation (1), around a closed path $a-b-c-d-a$, as shown in fig 4.4(a).

$$\oint \vec{E} \cdot d\vec{l} = \int_a^b \vec{E} \cdot d\vec{l} + \int_b^c \vec{E} \cdot d\vec{l} + \int_c^d \vec{E} \cdot d\vec{l} + \int_d^a \vec{E} \cdot d\vec{l} = 0$$

As the electric field inside a perfect conductor is zero, this means $\vec{E} = 0$. Thus, the above equation reduces to:

$$E_t \Delta w - E_n \frac{\Delta h}{2} - 0 \cdot \frac{\Delta h}{2} - 0 \Delta w + 0 \cdot \frac{\Delta h}{2} + E_n \frac{\Delta h}{2} = 0$$

$$E_t \Delta w = 0 \quad \text{Since } \Delta h \rightarrow 0$$

$$\boxed{E_t = 0} \quad \dots(10)$$

Similarly, to derive boundary conditions for normal components by applying eq. (2) to the pillbox of Figure 4.4(b) and letting $\Delta h \rightarrow 0$, we get

$$\Delta Q = D_n \cdot \Delta S - 0 \cdot \Delta S$$

because $\mathbf{D} = \epsilon \mathbf{E} = 0$ inside the conductor.

$$D_n = \frac{\Delta Q}{\Delta S} = \rho_s$$

or

$$\boxed{D_n = \rho_s} \quad (\dots 12)$$

Thus under static conditions, the following conclusions can be made about a perfect conductor:

1. No electric field may exist *within* a conductor; that is,

$$\boxed{\rho_v = 0, \quad \mathbf{E} = 0} \quad \dots(13)$$

2. Since $\mathbf{E} = -\nabla V = 0$, there can be no potential difference between any two points in the conductor; that is, a conductor is an equipotential body.
3. The electric field $\mathbf{E}$ can be external to the conductor and *normal* to its surface; that is

$$\boxed{D_t = \epsilon_0 \epsilon_r E_t = 0, \quad D_n = \epsilon_0 \epsilon_r E_n = \rho_s} \quad \dots(14)$$

C. Conductor–Free Space Boundary Conditions

This is a special case of the conductor–dielectric conditions and is illustrated in Figure 4.4 . The boundary conditions at the interface between a conductor and free space can be obtained from eq. (13) by replacing ϵ_r by 1 (because free space may be regarded as a special dielectric for which $\epsilon_r = 1$). We expect the electric field $\mathbf{E}$ to be external to the conductor and normal to its surface. Thus the boundary conditions are

$$D_t = \epsilon_0 E_t = 0, \quad D_n = \epsilon_0 E_n = \rho_s \quad \dots(14)$$

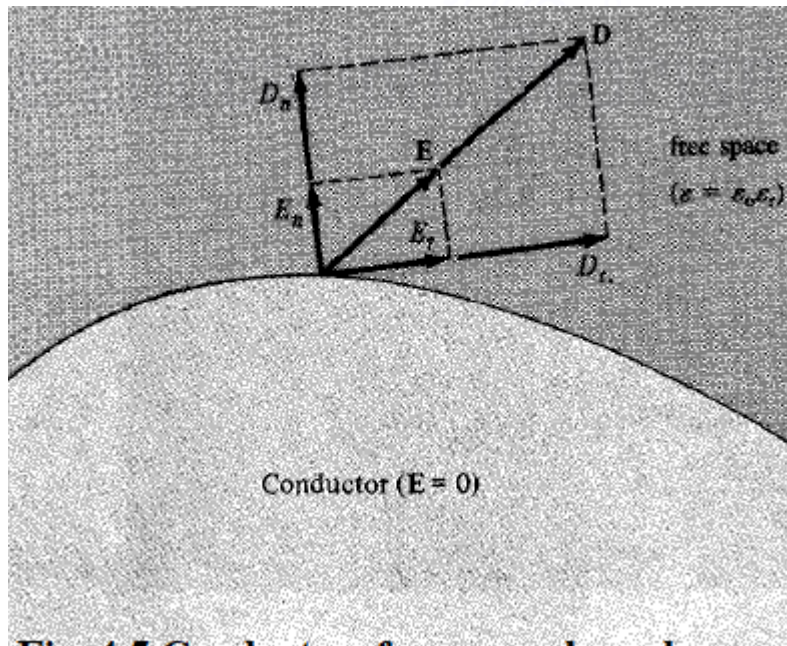


Fig. 4.5 Conductor -free space boundary

4.7 Boundary conditions for Magnetic fields

- ❖ To determine the Magnetic boundary conditions, we need to use Maxwell's equations:

$$\oint_l \vec{H} \cdot d\vec{l} = I_{enc} \quad \dots(15)$$

and

$$\oint_s \vec{B} \cdot d\vec{s} = 0 \quad \dots(16)$$

- ❖ The Magnetic field intensity $\vec{H}$ is decomposed into two orthogonal components:

$$\vec{H} = \vec{H}_t + \vec{H}_n \dots\dots(17)$$

where $\mathbf{H}_t$ and $\mathbf{H}_n$ are, respectively, the tangential and normal components of $\mathbf{H}$ to the interface of interest. A similar decomposition can be done for the magnetic flux density $\vec{B}$.

Consider two magnetic media 1 and media2 having magnetic permeabilites μ_1 and μ_2 , respectively.

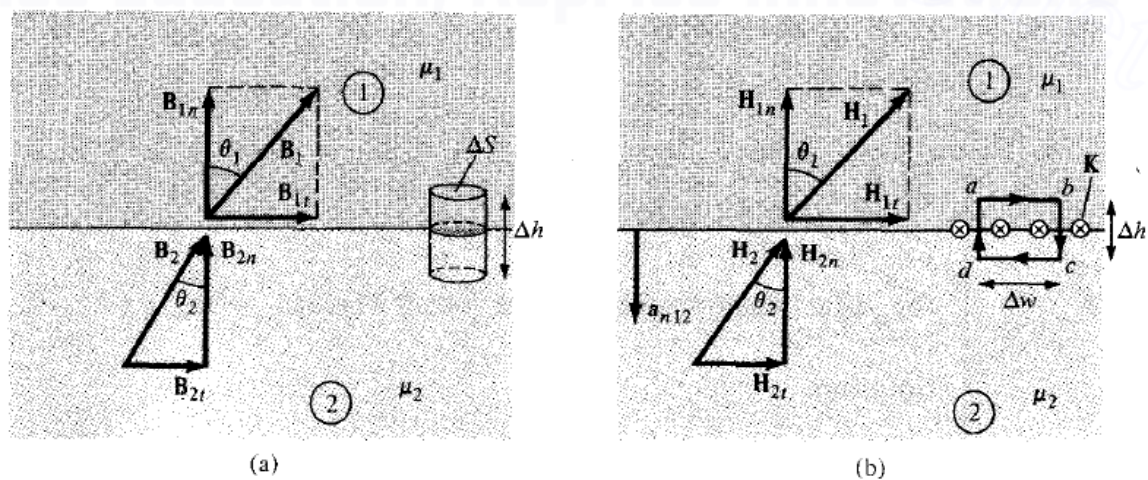


Figure 4.6 Boundary conditions between two magnetic media: (a) for $\vec{B}$, (b) for $\vec{H}$.

- To determine the **tangential components** for magnetic field $\mathbf{H}$, apply Ampere's law in eq(1) around the closed path a-b-c-d-a as shown in the fig4.6(b).

$$\oint \vec{H} \cdot d\vec{l} = I$$

Such that,

$$\oint \vec{H} \cdot d\vec{l} = \int_a^b \vec{H} \cdot d\vec{l} + \int_b^c \vec{H} \cdot d\vec{l} + \int_c^d \vec{H} \cdot d\vec{l} + \int_d^a \vec{H} \cdot d\vec{l} = K \Delta w$$

where $\vec{K}$ is surface current density, normal to the closed path.

$$H_{1t} \Delta w - H_{1n} \frac{\Delta h}{2} - H_{2n} \frac{\Delta h}{2} - H_{2t} \Delta w + H_{2n} \frac{\Delta h}{2} + H_{1n} \frac{\Delta h}{2} = K \Delta \quad \dots(18)$$

Now, as the surface approaches boundary, $\Delta h \rightarrow 0$ Thus, Equation (18) becomes:

$$H_{1t} - H_{2t} = K \quad \dots(19)$$

Or, it can be written as:

$$\frac{B_{1t}}{\mu_1} - \frac{B_{2t}}{\mu_2} = K$$

Or, in vector notation Equation (14) can be written as:

$$(\vec{H}_1 - \vec{H}_2) \times \hat{a}_{n_{12}} = \vec{K}$$

where $\hat{a}_{n_{12}}$ is the unit normal vector directed from material 1 to material 2.

If the boundary is free of current or the media are not conductors (for K is free current density), $K=0$ and eq. (19) becomes

$$H_{1t} = H_{2t} \quad \dots(20)$$

Or,

$$\frac{B_{1t}}{\mu_1} = \frac{B_{2t}}{\mu_2}$$

Thus, at the boundary of two materials, tangential component of $\vec{H}$ is continuous but the tangential component of $\vec{B}$ is discontinuous.

- To determine the **normal components** for magnetic field $\mathbf{B}$, apply Gauss's law in eq(2) to the pillbox(Gaussian surface) as shown in the fig4.6(a).

$$\oint \vec{B} \cdot d\vec{s} = 0$$

$$= \int_{top} \vec{B} \cdot d\vec{s} + \int_{bottom} \vec{B} \cdot d\vec{s} + \int_{sides} \vec{B} \cdot d\vec{s} = 0$$

and as surface approaches $\Delta h \rightarrow 0$, we obtain

$$B_{1n} \cdot \Delta s - B_{2n} \cdot \Delta s = 0 \quad \dots(21)$$

$$\boxed{B_{1n} = B_{2n}} \quad \dots(22)$$

Or,

$$\mu_1 H_{1n} = \mu_2 H_{2n}$$

Thus, at the boundary of two materials, normal component of $\vec{B}$ is continuous but the normal component of $\vec{H}$ is discontinuous.

UNIT-5

1. Derive the wave equation for Freespace? (VERY IMPORTANT)

5.2 Wave Equations for a free space (or perfect dielectric).

Consider a wave propagating in a free space medium. For free space, $\sigma = 0$, $\rho_v = 0$, $\epsilon = \epsilon_0$ and $\mu = \mu_0$. Maxwell's equations (I) to (IV) reduce for free space reduces to.

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \quad \text{--- (1)} \quad (\because J = \sigma \vec{E} = 0)$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{--- (2)}$$

$$\nabla \cdot \vec{D} = 0 \quad \text{--- (3)}$$

$$\nabla \cdot \vec{B} = 0 \quad \text{--- (4)}$$

Differentiating eq (1) with respect to time,

$$\begin{aligned} \frac{\partial}{\partial t} (\nabla \times \vec{H}) &= \frac{\partial^2 \vec{D}}{\partial t^2} \\ \Rightarrow \nabla \times \frac{\partial \vec{H}}{\partial t} &= \frac{\partial^2 \vec{D}}{\partial t^2} \quad \text{--- (5)} \end{aligned}$$

Also Since ϵ_0 and μ_0 are independent of time

$$\vec{D} = \epsilon_0 \vec{E} \Rightarrow \frac{\partial \vec{D}}{\partial t} = \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{--- (6)}$$

$$\vec{B} = \mu_0 \vec{H} \Rightarrow \frac{\partial \vec{B}}{\partial t} = \mu_0 \frac{\partial \vec{H}}{\partial t} \quad \text{--- (7)}$$

$\therefore$ eq (5) becomes

$$\nabla \times \frac{\partial \vec{H}}{\partial t} = \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{--- (8)}$$

Taking curl of both sides of (2) and using (7), we obtain

$$\nabla \times \nabla \times \vec{E} = -\mu_0 \nabla \times \frac{\partial \vec{H}}{\partial t} = -\mu_0 \nabla \times \frac{\partial \vec{H}}{\partial t}$$

$$\Rightarrow \nabla \times \nabla \times \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{--- (9)}$$

from vector identity

$$\nabla \times \nabla \times \vec{E} = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$$

combining this equation with (9)

$$\nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \quad \text{--- (10)}$$

But

$$\nabla \cdot \vec{E} = \frac{1}{\epsilon_0} (\nabla \cdot \vec{D}) = 0$$

$\therefore$ eq (10) becomes

$$-\nabla^2 \vec{E} = -\mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\Rightarrow \boxed{\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}} \quad \text{--- (11)}$$

$$\text{Ily} \quad \boxed{\nabla^2 \vec{H} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2}} \quad \text{--- (12)}$$

Equation (11) and (12) are known as wave equation for free space.

The wave velocity in free space is $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$.

2. For a conducting medium derive expressions for α and β .

7.10 Expressions for α and β for a Conducting Medium

Consider a wave propagating in a lossy conducting medium. The propagation constant is

$$\begin{aligned}\gamma &= \alpha + j\beta \\ &= \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}.\end{aligned}$$

$$\begin{aligned}\gamma^2 &= (\alpha + j\beta)^2 = j\omega\mu(\sigma + j\omega\epsilon), \\ \alpha^2 - \beta^2 + 2j\alpha\beta &= -\omega^2\mu\epsilon + j\omega\mu\sigma.\end{aligned}\tag{7.63}$$

Equating real and imaginary terms,

$$\alpha^2 - \beta^2 = -\omega^2\mu\epsilon\tag{7.64}$$

and $2\alpha\beta = \omega\mu\sigma$.

Since $(a+b)^2 = (a-b)^2 + 4ab$,

$$(\alpha^2 + \beta^2)^2 = (\alpha^2 - \beta^2)^2 + 4\alpha^2\beta^2.$$

$$\text{or } (\alpha^2 + \beta^2)^2 = (-\omega^2\mu\epsilon)^2 + (\omega\mu\sigma)^2.$$

$$\therefore \alpha^2 + \beta^2 = \sqrt{\omega^4\mu^2\epsilon^2 + \omega^2\mu^2\sigma^2} = \omega\mu\sqrt{\omega^2\epsilon^2 + \sigma^2}.\tag{7.65}$$

Solving Eq. (7.64) and Eq. (7.65),

$$2\alpha^2 = \omega\mu\sqrt{\omega^2\epsilon^2 + \sigma^2} - \omega^2\mu\epsilon.$$

$$= \omega^2\mu\epsilon\sqrt{1 + \frac{\sigma^2}{\omega^2\epsilon^2}} - \omega^2\mu\epsilon$$

$$\text{or } \alpha = \omega\sqrt{\frac{\mu\epsilon}{2}\left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} - 1\right]}\tag{7.66}$$

$$\text{and } 2\beta^2 = \omega\mu\sqrt{\omega^2\epsilon^2 + \sigma^2} + \omega^2\mu\epsilon.$$

$$\beta = \omega\sqrt{\frac{\mu\epsilon}{2}\left[\sqrt{1 + \left(\frac{\sigma}{\omega\epsilon}\right)^2} + 1\right]}.\tag{7.67}$$

3. Write short notes on the following
 (a) Brewster angle (b) Total Internal Reflection.

BREWSTER ANGLE

Definition Brewster angle is the angle of incidence at which there is no reflection.

Brewster angle for parallel polarisation

For parallel polarisation, we have

$$\frac{E_r}{E_i} = \frac{\frac{\epsilon_{r2}}{\epsilon_{r1}} \cos \theta_i - \sqrt{\left(\frac{\epsilon_{r2}}{\epsilon_{r1}}\right)^2 - \sin^2 \theta_i}}{\frac{\epsilon_{r2}}{\epsilon_{r1}} \cos \theta_i + \sqrt{\left(\frac{\epsilon_{r2}}{\epsilon_{r1}}\right)^2 - \sin^2 \theta_i}}$$

At Brewster angle, $\frac{E_r}{E_i} = 0$

$$\frac{\epsilon_{r2}}{\epsilon_{r1}} \cos \theta_i = \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}} - \sin^2 \theta_i}$$

$$\left(\frac{\epsilon_{r2}}{\epsilon_{r1}}\right)^2 \cos^2 \theta_i = \frac{\epsilon_{r2}}{\epsilon_{r1}} - \sin^2 \theta_i$$

$$\left(\frac{\epsilon_{r2}}{\epsilon_{r1}}\right)^2 - \left(\frac{\epsilon_{r2}}{\epsilon_{r1}}\right)^2 \sin^2 \theta_i = \frac{\epsilon_{r2}}{\epsilon_{r1}} - \sin^2 \theta_i$$

Simplifying this, we get

$$\sin^2 \theta_i = \frac{\epsilon_{r2}}{\epsilon_{r1} + \epsilon_{r2}}$$

or,

$$\cos^2 \theta_i = \frac{\epsilon_{r2}}{\epsilon_{r1} + \epsilon_{r2}} - 1 = \frac{\epsilon_{r1}}{\epsilon_{r1} + \epsilon_{r2}}$$

$$\tan \theta_i = \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}}$$

Brewster angle,

$$\theta_i = \tan^{-1} \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}}$$

TOTAL INTERNAL REFLECTION

Definition Total internal reflection is said to exist if

1. the angle of incidence is very high
2. medium 1 is denser than medium 2 and

3. $\sin \theta_i > \sqrt{\frac{\epsilon_{r2}}{\epsilon_{r1}}}$

For total internal reflection,

$$\frac{E_r}{E_i} = \frac{\cos \theta_i + j \sqrt{\sin^2 \theta_i - \left(\frac{\epsilon_2}{\epsilon_1}\right)}}{\cos \theta_i - j \sqrt{\sin^2 \theta_i - \left(\frac{\epsilon_2}{\epsilon_1}\right)}} \text{ for perpendicular polarisation}$$

$$\frac{E_r}{E_i} = \frac{\left(\frac{\epsilon_2}{\epsilon_1}\right) \cos \theta_i + j \sqrt{\sin^2 \theta_i - \left(\frac{\epsilon_2}{\epsilon_1}\right)}}{\left(\frac{\epsilon_2}{\epsilon_1}\right) \cos \theta_i - j \sqrt{\sin^2 \theta_i - \left(\frac{\epsilon_2}{\epsilon_1}\right)}} \text{ for parallel polarisation}$$

The concept of total internal reflection is often used in binocular optics. For these applications, glass prisms are used to shorten the instrument.

4. Find α , β , ν and η for Ferrite at 10GHz $\epsilon_r = 9$, $\mu_r = 4$, $s = 10\text{ms/m}$.

Example 7.11 Find α , β , γ and η for ferrite at 10 GHz when $\epsilon_r = 9$, $\mu_r = 4$, $\sigma = 10 \text{ mS/m}$.

Solution Given: $\epsilon_r = 9$, $\mu_r = 4$, $f = 10 \times 10^9 \text{ Hz}$ and $\sigma = 10 \times 10^{-3} \text{ mS/m}$.

Since $\sigma \ll \omega\epsilon$, for a dielectric we know that,

$$\alpha = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} = \frac{10 \times 10^{-3}}{2} \sqrt{\frac{4\mu_0}{9\epsilon_0}} = 1.256 \text{ Np/m.}$$

$$\beta = \omega \sqrt{\mu \epsilon} \left(1 + \frac{\sigma^2}{8\omega^2 \epsilon^2} \right)$$

$$= 2\pi \times 10^{10} \sqrt{4\mu_0 \times 9\epsilon_0} \left(1 + \frac{10^{-4}}{8 \times 0.556^2} \right) = 1256.6 \text{ rad/m.}$$

$$\text{Velocity of propagation, } v_p = \frac{\omega}{\beta} = \frac{2\pi f}{\beta} = \frac{2\pi \times 10^{10}}{1256.6} = 0.5 \times 10^8 \text{ m/s.}$$

$$\begin{aligned} \text{Intrinsic impedance } \eta &= \sqrt{\frac{\mu}{\epsilon}} \left(1 + j \frac{\sigma}{2\omega\epsilon} \right) \\ &= \sqrt{\frac{4\mu_0}{9\epsilon_0}} \left(1 + j \frac{10^{-2}}{2 \times 0.556} \right) = 251.32 (1 + j9 \times 10^{-3}). \end{aligned}$$

$$\therefore \eta = 251.33 \angle 0.52^\circ \Omega.$$

5. For good dielectrics derive the expressions for α , β , v and η

7.12 Wave Propagation in Good Dielectrics

Consider a wave propagating in a good dielectric medium. For good dielectrics, the conductivity is very low compared to $\omega\epsilon$, i.e., $\sigma \ll \omega\epsilon$.

The expression for the propagation constant is

$$\begin{aligned} \gamma &= \sqrt{j\omega\mu(\sigma + j\omega\epsilon)} \\ &= (\sqrt{j\omega\mu})(\sqrt{j\omega\epsilon}) \sqrt{1 + \frac{\sigma}{j\omega\epsilon}}. \\ \gamma &= j\omega\sqrt{\mu\epsilon} \sqrt{1 + \frac{\sigma}{j\omega\epsilon}}. \end{aligned} \tag{7.72}$$

Now, using binomial expansion,

$$\sqrt{1 + \frac{\sigma}{j\omega\epsilon}} \approx 1 + \frac{\sigma}{2j\omega\epsilon} + \left(\frac{-1}{4} \right) \frac{\sigma^2}{2j^2\omega^2\epsilon^2} + \dots$$

Since $\frac{\sigma}{\omega\epsilon} \ll 1$, higher order terms can be neglected

Therefore a good approximation is

$$\sqrt{1 + \frac{\sigma}{j\omega\epsilon}} \approx 1 + \frac{\sigma}{2j\omega\epsilon} + \frac{\sigma^2}{8\omega^2\epsilon^2}. \quad (7.73)$$

Substituting this value in Eq. (7.72),

$$\gamma = j\omega\sqrt{\mu\epsilon} \left[1 + \frac{\sigma}{2j\omega\epsilon} + \frac{\sigma^2}{8\omega^2\epsilon^2} \right].$$

$$\alpha + j\beta = j\omega\sqrt{\mu\epsilon} + \frac{j\omega\sqrt{\mu\epsilon}\sigma}{2j\omega\epsilon} + \frac{j\omega\sqrt{\mu\epsilon}\sigma^2}{8\omega^2\epsilon^2},$$

$$\text{or } \alpha + j\beta = j\omega\sqrt{\mu\epsilon} + \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} + \frac{j\sigma^2\sqrt{\mu\epsilon}}{8\omega^2\epsilon^2}. \quad (7.74)$$

Equating real and imaginary terms, the attenuation constant is

$$\alpha = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} \quad (7.75)$$

and the phase shift constant is

$$\beta = \omega\sqrt{\mu\epsilon} \left(1 + \frac{\sigma^2}{8\omega^2\epsilon^2} \right), \quad (7.76)$$

Note: For a perfect or lossless dielectric, $\sigma = 0$, and $\beta = \omega\sqrt{\mu\epsilon}$.

Intrinsic impedance We know that the intrinsic impedance for a lossy medium is

$$\begin{aligned} \eta &= \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}} \\ &= \sqrt{\frac{\mu}{\epsilon}} \sqrt{\frac{1}{\left(1 + \frac{\sigma}{j\omega\epsilon}\right)}} = \sqrt{\frac{\mu}{\epsilon}} \left(1 + \frac{\sigma}{j\omega\epsilon}\right)^{-1/2}. \end{aligned}$$

Using the binomial theorem expansion,

$$\eta = \sqrt{\frac{\mu}{\epsilon}} \left(1 - \frac{\sigma}{2j\omega\epsilon} + \dots \right).$$

Neglecting higher order terms, the intrinsic impedance for good dielectrics is

$$\eta = \sqrt{\frac{\mu}{\epsilon}} \left(1 + j\frac{\sigma}{2\omega\epsilon} \right). \quad (7.77)$$

Thus, when a wave is travelling in a lossy dielectric medium, due to conductivity a reactive component is added in the impedance.

Phase velocity We know that the velocity of the wave is

$$v_p = \frac{\omega}{\beta} = \frac{\omega}{\omega \sqrt{\mu \epsilon} \left(1 + \frac{\sigma^2}{8\omega^2 \epsilon^2} \right)}$$

$$v_p = \frac{1}{\sqrt{\mu \epsilon} \left(1 + \frac{\sigma^2}{8\omega^2 \epsilon^2} \right)^{-1}}$$

Now, using binomial expansion and neglecting higher order terms, phase velocity in good dielectrics is

$$v_p = \frac{v_0}{\sqrt{\mu_r \epsilon_r} \left(1 - \frac{\sigma^2}{8\omega^2 \epsilon^2} \right)} \quad (7.78)$$

6. . Show that for a good conductor, the surface impedance is equal to the intrinsic impedance

The surface impedance is

$$Z_s = \frac{E_{\tan}}{J_s} = \frac{\gamma}{\sigma} \Omega. \quad (\text{Proved})$$

We know that the propagation constant is $\gamma = \sqrt{j\omega\mu(\sigma + j\omega\epsilon)}$.

$$\therefore Z_s = \frac{\sqrt{j\omega\mu(\sigma + j\omega\epsilon)}}{\sigma} \Omega. \quad (8.65)$$

Thus the surface impedance depends on the conductor properties, ϵ , μ , σ at a given frequency.

Note: For a good conductor, $\sigma \gg \omega\epsilon$.

$$\therefore \gamma = \sqrt{j\omega\mu\sigma}.$$

$$\text{Surface impedance is } Z_s = \frac{\sqrt{j\omega\mu\sigma}}{\sigma} = \sqrt{\frac{\omega\mu}{\sigma}} \angle 45^\circ.$$

$$\text{We know that the characteristic impedance } \eta = \sqrt{\frac{\omega\mu}{\sigma}} \angle 45^\circ. \quad (8.66)$$

CS Scanned with CamScanner Therefore, for a good conductor, the surface impedance is equal to the characteristic impedance.

7. Describe the concept of Reflection of an EM wave by a perfect dielectric at oblique incidence.

8.8 Reflection by a Dielectric or Perfect Insulator—Oblique Incidence

Consider a plane wave incident obliquely on the surface of a dielectric medium or a perfect insulator. Some part of the wave is reflected and the other part is refracted through the medium at the boundary as shown in Fig. 8.8.

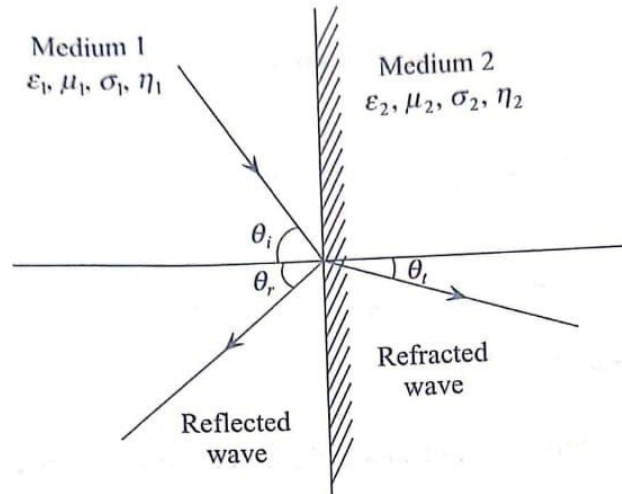


Fig. 8.8 Oblique incidence of a plane wave on a dielectric medium

Let θ_i = incident angle, θ_r = reflected angle and θ_t = refracted angle. $\theta_i = \theta_r = \theta_1$, $\theta_t = \theta_2$..
 Also let medium 1 and 2 have the properties ϵ_1 , η_1 , β_1 and ϵ_2 , η_2 , β_2 respectively.
 From Snell's law for dielectric media,

$$\sqrt{\epsilon_1} \sin \theta_1 = \sqrt{\epsilon_2} \sin \theta_2. \quad (8.38)$$

The field vectors can be obtained from the distribution of powers at the boundary surface.

Assume that the power absorbed by the material is neglected. We know that the wave power density is

$$P = |\vec{E} \times \vec{H}| = \frac{E^2}{\eta}. \quad (8.39)$$

Incident wave power density is

$$P_i = \frac{E_i^2}{\eta_1} \cos \theta_1. \quad (8.40)$$

Similarly, the reflected power density is

$$P_r = \frac{E_r^2}{\eta_1} \cos \theta_1 \quad (8.41)$$

and the transmitted wave power density is

$$P_t = \frac{E_t^2}{\eta_2} \cos \theta_2. \quad (8.42)$$

From the principle of conservation of energy, we know that

Incident power density = reflected power density + refracted power density

i.e., $P_i = P_r + P_t$.

$$\frac{E_i^2}{\eta_1} \cos \theta_1 = \frac{E_r^2}{\eta_1} \cos \theta_1 + \frac{E_t^2}{\eta_2} \cos \theta_2.$$

$$1 = \frac{E_r^2}{E_i^2} + \frac{E_t^2}{E_i^2} \left(\frac{\eta_1}{\eta_2} \right) \frac{\cos \theta_2}{\cos \theta_1}.$$

We know that $\frac{\eta_1}{\eta_2} = \frac{\sqrt{\mu_0/\epsilon_1}}{\sqrt{\mu_0/\epsilon_2}} = \sqrt{\frac{\epsilon_2}{\epsilon_1}}.$

$$\therefore \frac{E_r^2}{E_i^2} = 1 - \frac{E_t^2}{E_i^2} \sqrt{\frac{\epsilon_2}{\epsilon_1}} \left(\frac{\cos \theta_2}{\cos \theta_1} \right), \quad (8.43)$$

where $\frac{E_r}{E_i} = \Gamma$, is the reflection coefficient and $\frac{E_t}{E_i} = T$, is the transmission coefficient.

To find the reflection and transmission coefficients, the wave polarizations should be considered.



BONAM VENKATA CHALAMAYYA ENGINEERING COLLEGE
(AUTONOMOUS)
ODALAREVU-533210

EMWTL QUESTION BANK

UNIT-1

1.a) Draw the equivalent circuit for two-wire transmission line and also obtain transmission line equations of voltage and current.

(OR)

Derive the relationship between secondary constant and primary constants of a transmission line.

b) For lossless line, $Z_0 = 50 \Omega$ and $u = 2.8 \times 10^8$ m/s. Determine L and C for the line.

2.a) List out different types of transmission lines and write their applications.

b) A high frequency line has the following primary constants $L=1.2\text{mH/Km}$, $C=0.05\mu\text{F/Km}$. $R = G = \text{negligible}$. Determine the characteristic impedance and propagation constant of the line.

3. Sketch the T-type and π -type equivalent circuit of Transmission line

4. Define characteristic impedance of a line and Derive its expression.

5. Explain the concept of infinite line.

6. Define Wavelength, phase velocity and group velocity. Derive the relationship between phase velocity(V_g) and Group velocity(V_p).

7.a) Discuss the conditions for lossy, loss less and distortion less transmission lines

b) For lossless line, $Z_0 = 50 \Omega$ and $u = 2.8 \times 10^8$ m/s. Determine L and C for the line.

UNIT-II

1. Define input impedance of a transmission line and derive the input impedance for OC and SC lines.

2. a) Define Reflection coefficient and derive the expression for input impedance in terms of reflection coefficient.

b) Derive the relationship between reflection coefficient and VSWR.

3. Explain the significance and utility of $\lambda/8$, $\lambda/4$, and $\lambda/2$ lines

4. Explain UHF lines as circuit elements with equivalent circuits.

5. Define Smith chart? Explain the construction and properties of the smith chart and its Applications

6. Explain the design procedure of (i) Single stub matching and ii) Double Stub matching

7. Explain the operation of quarter wave transformer and mention its applications



BONAM VENKATA CHALAMAYYA ENGINEERING COLLEGE
(AUTONOMOUS)
ODALAREVU-533210

UNIT-III

1. State and explain Coulomb's law. Obtain an expression in vector form.
2. Derive the expression for electric field intensity due to infinite line charge and surface charge.
3. State and prove the Gauss's law in electrostatics (Maxwells's equation) and Explain its applications.
4. Define Linear, isotropic and homogeneous dielectrics
5. What are poisson's and Laplace equations, and explain the significance
6. State Biot-savart's law? Derive the 'H' field at an observation point due to infinite long current element?
7. State and Prove the Ampere's Law(**Maxwell's equation in magnetostatics**) and Explain its applications.
8. Write about Magnetic Vector and Scalar Potentials.
9. Write the Maxwell's equations of EM static field in point form and integral form?
10. Write a short notes on
 - i) Equation of continuity for time varying fields.
 - ii) Relaxation time

UNIT-IV

1. What is Faraday's Law for induced emf and Explain Transformer emf and Motional emf.
2. Derive the boundary conditions for the tangential and normal components of Electrostatic fields at the boundary between i) Dielectric-Dielectric ii) Conductor-Dielectric
3. Prove that H_{tan} is discontinuous and B_{norm} is continuous at boundary between two media?
4. What is inconsistency associated with Ampere's law ? How is it rectified?
5. Derive Maxwell's equations in Integral and Differential forms for time varying fields and Write their word statement.
6. Derive Maxwell's Equations for EM fields varying harmonically with time.
7. Write the difference between Conduction Current density and displacement current density.

UNIT-V

1. What is a uniform plane waves and explain the its properties
2. Prove thae E and H waves are perpendicular to each other.
3. Derive the relation between E and H in uniform plane wave propagation
4. Obtain the wave equations for free space and conducting medium.



BONAM VENKATA CHALAMAYYA ENGINEERING COLLEGE
(AUTONOMOUS)
ODALAREVU-533210

5. Derive the expressions for attenuation constant(α) and phase constant(β) for conducting medium.
6. For good conductors and good dielectrics derive the expressions for $\alpha, \beta, \gamma, V_p$ and η .
7. State and prove Poynting theorem.
8. Explain the different types of polarization of uniform plane wave
9. Derive the expression for Reflection and Transmission coefficients of an EM wave when it is incident normally on a dielectric and perfect conductor.
10. Write short notes on (i) Brewster Angle and (ii) Total Internal reflection
11. Define and explain the terms: i) skin depth ii) intrinsic impedance of free space
iii) Surface Impedance